Title: Upslope range shifts are robust to relocation error in a historical vegetation resurvey

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Abstract (250 words max):

Climate change is reshaping species' geographic ranges worldwide, particularly in mountain ecosystems, where rapid warming is driving species to track suitable environmental conditions upslope. However, range shifts vary among species and may be delayed or difficult to detect. Disentangling biological signal from noise is therefore important for understanding this variation. Resurveys are powerful for detecting temporal changes in species distributions and abundance, but their interpretation can be complicated by methodological uncertainty, particularly plot relocation, a source of error rarely quantified in resurvey studies. We resurveyed 43 plots (17,200 m²) in E.C. Manning Provincial Park, Canada, to assess temporal changes in the elevational distributions of montane plants while accounting for plot relocation error. We quantified relocation error by resurveying nine plots, both at their original locations and nearby locations that simulated plausible relocation uncertainty. By comparing turnover between these paired plots, we developed informative priors to account for relocation error in hierarchical Bayesian models of species occurrence across elevation. Across 30 common species, models predicted upslope shifts despite accounting for relocation error. However, responses differed among functional groups: herbs showed the strongest evidence of upslope shifts, shrubs increased in probability of occurrence without a directional elevational shift, and trees showed no evidence of change in either measure. Our findings reveal contrasting elevational responses among plant functional groups and demonstrate an approach for incorporating empirically estimated relocation error into resurvey studies. This framework can strengthen inference from historical vegetation resurveys and support future assessments of biodiversity responses to climate change.

Keywords: Climate change, Vegetation resurvey, Elevational range shift, Relocation error, Pseudoturnover, Bayesian inference, Mountain ecosystem, Temperate forest, Temporal vegetation change

Introduction

Climate change is transforming ecosystems, altering community composition, structure, and functioning globally (Dainese et al., 2024; Mamantov et al., 2021; Parmesan, 2006). With the current rate of climate change exceeding the capacity for acclimatization and evolutionary adaptation in many species, particularly long-lived species, the ability to shift ranges becomes increasingly important for persistence under changing conditions (Burrows et al., 2011; Hoffmann & Sgrò, 2011; Jump & Peñuelas, 2005). Because temperature is an important determinant of a species' potential range (Pearson & Dawson, 2003; Woodward & Williams, 1987), as the climate warms globally, species are expected to shift both polewards in latitude and

upwards in elevation to track suitable habitat (Loarie et al., 2009; Parmesan & Yohe, 2003). While this is a well-documented phenomenon (Chen et al., 2011; Mamantov et al., 2021), recent reviews have highlighted great variability in the magnitude and direction of range shifts across species, geographic regions, and studies (Lawlor et al., 2024; Rubenstein et al., 2023). Deviations from expectations can be attributed to several factors, including species' intrinsic sensitivities and capacities to track climate, overriding effects of non-climatic (i.e. biotic, edaphic) factors, and methodological challenges when detecting range shifts (Shih Mendez et al., 2026). Correctly attributing range shifts to climate change requires careful consideration of both the biological characteristics of species and the methodological processes through which range shifts are detected.

Some variation in the rate and direction of range shifts can be attributed to differences in species' capacity to track changing climates due to differences in life history, dispersal ability, stress tolerance, and functional traits. Species with shorter generation times and greater reproductive or dispersal capacity may respond more rapidly to changing environmental conditions than long-lived species with slower demographic rates (Comte et al., 2024; Couet et al., 2022; Lenoir et al., 2008). In mountain systems, for instance, herbaceous species have been observed to shift upslope faster and farther than woody, slow-growing species (Lenoir et al., 2008). However, range shifts do not always follow the direction expected from warming alone, as changes in non-thermal climatic and environmental drivers can alter habitat suitability across elevations. Changes in precipitation regimes have been documented to make lower elevations more favourable, leading to downslope shifts in species distributions (Lenoir et al., 2010). On the other hand, sometimes species may resist shifting and persist within their historical ranges when local conditions buffer exposure to regional warming or plasticity increases tolerance to it. Microhabitat refugia, such as those created beneath dense tree canopies, can facilitate microclimate buffering by maintaining cooler, more stable conditions for persistence (De Frenne et al., 2013), while phenotypic plasticity may allow species to acclimatize to changing conditions and delay the need for immediate range shifts (Jump & Peñuelas, 2005). Conversely, species may lag behind when slow colonization or delayed mortality constrain their ability to track climate (Goodwin et al., 2025), and can be further constrained by biotic interactions, such as competition, that limit establishment in newly suitable habitats (Bjorkman et al., 2017). As a result, these mechanisms can decouple observed species distributions from the spatial trajectory expected under a changing climate.

However, variation in detected range shifts may not arise solely from differences in species' responses to climate change; it can also result from methodological uncertainty inherent to empirical estimates of range shifts. Resurvey studies of historical plots established years or decades earlier are commonly used to detect climate-driven range shifts and are particularly valuable for overcoming the temporal limitations of short-term studies (Vellend et al., 2013). However, historical plots are often semi-permanent or unmarked, and their original locations may be difficult to precisely relocate, especially for historical locations recorded on hand-drawn

maps or compass bearings before the widespread use of handheld GPS (Kapfer et al., 2016). Despite improved GPS location accuracy in recent decades (Verma & Tracer, 2025), when present-day resurveys do not precisely match their historical locations, differences in species composition could reflect spatial heterogeneity rather than true temporal change, producing what is known as pseudoturnover (Fischer & Stöcklin, 1997). To address this, some studies rely on landscape and vegetation information to relocate plots by matching the elevation, slope, and aspect to those of the original plot (Kopecký & Macek, 2015), or by matching tree canopy composition, given the long lifespan of mature trees (Rogers et al., 2008). Although commonly acknowledged as a potential source of methodological error in resurvey studies, the effects of relocation error are often overlooked and rarely quantified. For studies that have explicitly tested the robustness of resurveys to relocation error, the results are mixed (Boch et al., 2022; Kopecký & Macek, 2015; Ross et al., 2010; Shih Mendez et al., 2026; Verheyen et al., 2018), suggesting that historical resurveys may or may not be robust to relocation error depending on factors such as plot size and replication. Despite these mixed results, general consensus exists that resurvey studies should be interpreted cautiously and sources of methodological error should be quantified as much as possible (Shih Mendez et al. 2026, Verheyen et al. 2018).

A potentially useful way to address relocation error is through hierarchical Bayesian modelling. Hierarchical Bayesian models are often used to incorporate measurement error into ecological analyses, allowing uncertainty in the observation process to be separated from variation in the underlying ecological process (Brusa et al., 2021; Park et al., 2026; Zou et al., 2023). Specifically, information from independent measurements or error assessments can be used to inform the model's prior distributions, allowing uncertainty associated with a measurement process to be incorporated into subsequent model estimates (Zou et al., 2023). In this study, relocation error is first quantified empirically by comparing species occurrence between accurately relocated and inaccurately relocated plots. This information is then used to construct informative priors for a model of temporal changes in elevational distributions, allowing uncertainty attributable to relocation error to be propagated to estimates of range shifts. To our knowledge, this is the first study to quantitatively incorporate relocation error into modeled estimates of climate-driven range shifts.

Our research objective was to determine whether montane plant species have shifted their elevational distributions over time while explicitly accounting for uncertainty associated with plot relocation error. We resurveyed 43 historical vegetation plots (20 x 20 m plots-17,200 m² total area) across E.C. Manning Provincial Park, British Columbia, Canada, 34 of which were originally surveyed between 1976 and 2000. To quantify relocation error, we resurveyed nine plots from 2019 twice in 2025, once at their precise historical coordinates (i.e., with no relocation error) and once at a nearby location with similar elevation, slope, aspect, and tree canopy composition (i.e., simulating typical relocation uncertainty for older plots). Differences in species occurrence between these paired resurveys were used to quantify variation associated with relocation error and inform priors in a Bayesian model of temporal changes in elevational

distributions for the remaining 34 resurvey plots. Our research questions were as follows: (1) Have montane plant species shifted their elevational distributions over time, and specifically, have they shifted upslope as expected under climate warming? We hypothesized warming has driven upslope range shifts across the park's elevational gradient as species redistribute to track suitable climatic conditions. (2) How does incorporating empirically quantified relocation error affect estimates of these range shifts? We assessed whether accounting for relocation error altered estimates of range shifts. (3) Do observed range shifts differ among functional groups (i.e. herbs, shrubs, and trees)? We hypothesized herbaceous species would exhibit greater upslope range shifts than woody shrub and tree species, given their generally shorter generation times and greater capacity for rapid demographic responses.

Methods

Study region

During the months of May to August 2025, 43 semi-permanent historical vegetation plots, originally sampled in 1976, 1979, 1993, 2000, and 2019 were resurveyed across E.C. Manning Provincial Park (Figure 1). E.C. Manning Provincial Park is located in a mountainous region in the southern interior of British Columbia, Canada, between the Okanagan and coast-interior transitional zones. With plots spanning an elevation gradient of 680– 2100 m, it is an ideal system for examining elevational trends and vegetation responses to climate change. Mountainous regions are among systems most sensitive to climate change, as the rate of temperature change is amplified due to higher altitudes (Pepin et al., 2015; Rangwala & Miller, 2012), causing shifts in seasonal weather patterns and reductions in snow cover (IPCC, 2019). Over the last five decades alone, the mean annual temperature at our study plots in Manning Park has risen by approximately 1.53 °C (ClimateNA, 2024, Fig S1), and warming across British Columbia has been most pronounced in the winter, leading to earlier snowmelt, longer growing seasons, and changing moisture regimes (British Columbia Ministry of Environment, 2016).

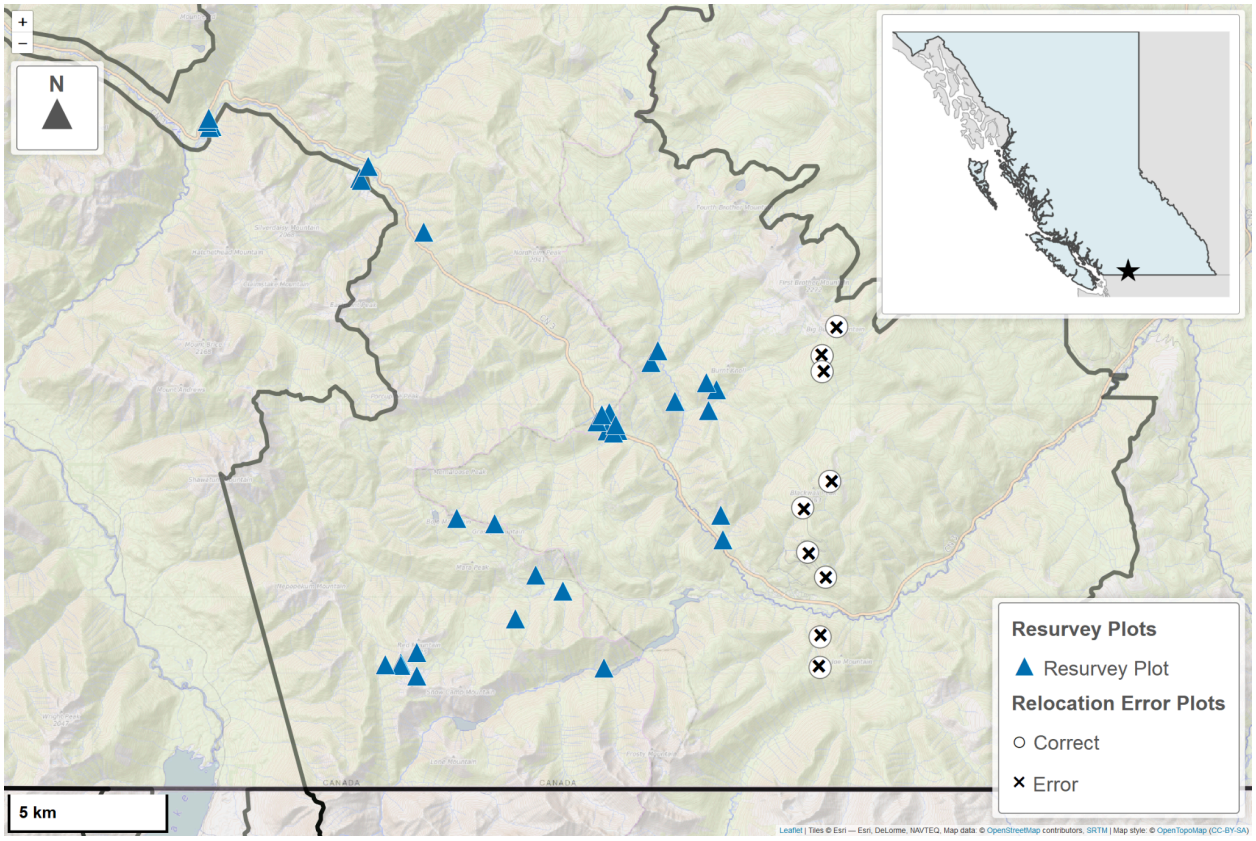


Figure 1. Locations of vegetation resurvey plots and relocation error plots within E.C. Manning Provincial Park, British Columbia, Canada. Blue triangles indicate resurveyed historical plots. Relocation error was assessed at separate plots, with open circles indicating correctly relocated plots ($2025_{Correct}$) and black crosses indicating error plots (2025_{Error}). The park boundary is overlaid on a topographic basemap derived from OpenStreetMap and OpenTopoMap, created in R using the leaflet package. The inset shows the location of E.C. Manning Provincial Park within British Columbia as a star.

Study design

We estimated elevational range shifts between historical vegetation surveys and 2025 resurveys using Bayesian mixed-effects models. Because historical plots are semi-permanent, uncertainty in their original locations can introduce apparent changes in species occurrence that are unrelated to ecological change. We therefore quantified the effects of plausible plot relocation error using paired resurveys of accurately and inaccurately relocated plots (*see Plot relocation protocol*), and quantified this uncertainty into Bayesian priors, which were then incorporated into the full model of elevational change. Posterior predictions from the full model were used to estimate shifts in species' elevational optima between the historical and present periods. We extended the same framework to examine whether range shifts differed among herbs, shrubs, and trees, with the prediction that herbaceous species would exhibit a greater upslope shift through time.

Historical data

Historical data were obtained from British Columbia's Ministry of Forests Biogeoclimatic Ecosystem Classification (BEC) database (W. H. MacKenzie, Accessed April 2024). This database contains historical vegetation surveys spanning the province of British Columbia, Canada, dating as far back as the early 1950s, conducted to map ecosystems across the province. It contains over 70,000 plots sampled using a standardized protocol (British Columbia Ministry of Forests & British Columbia Ministry of Environment, 2010 (ver. 2023))

We first subset the BEC database to plots within our area of interest in E.C. Manning Park (49.0–49.4° N, 120.5–121.0° W). As the plot data from our area of interest spanned multiple decades, data from 1976 to 2000 were combined and treated as the historical baseline for change. This also accounted for unevenness of sampling across elevations that was inherent in the data, as lower elevation plots tended to be sampled in earlier years (Figure S2). The data used in this study were also collected under several different projects and surveyors over time, so to ensure estimates of change were not biased by differences in sampling effort, species richness (i.e., the number of species detected per plot), was compared across project groups (*see Supplementary S3*). Richness did not differ significantly across most historical survey projects or between historical surveys and 2025 resurveys, indicating that sampling effort was comparable and unlikely to systematically influence species detections. To further reduce observer and potential sampling error, to ensure there was sufficient data for each species, and reduce the influence of rare species in modelling, analyses were restricted to the 30 most common species observed across historical and 2025 survey data. Species were only retained if they had a combined minimum of 20 occurrences across both survey periods and at least 7 occurrences in one of the two periods.

Data collection

Plot relocation protocol

Historical plot centers were initially relocated using Global Positioning System (GPS) coordinates recorded in the BEC database and a Garmin inReach 66i device. Latitude, longitude, and elevation for each plot were obtained from the BEC database, along with descriptive plot features recorded by the original surveyors. Because latitude and longitude were provided for plots surveyed before 1989, when handheld GPS devices became commercially available, it is likely that earlier coordinates were translated from mapped locations and later entered into the database. For these reasons, plot locations surveyed prior to 1989 may have lower GPS coordinate accuracy. Even for plots surveyed after this period, older GPS units tend to be less accurate, meaning coordinates from earlier survey years are generally less precise (Verma & Tracer, 2025). For example, reported coordinate accuracy in the BEC database improved from approximately 100 m in earlier surveys (~2000) to 7 m in more recent surveys (2025).

Given these limitations, plot centers were not relocated based on GPS coordinates alone. Upon arrival at the coordinates provided, we made informed plot relocation decisions based on all plot descriptions provided by the original surveyors. Often, plot centers were shifted to better match plot descriptions such as percent slope, aspect, tree canopy composition, and surveyor notes. In a few instances, plots were relocated several hundred meters from the coordinates provided based on exact relocation directions left by the original surveyors. To measure our own relocation confidence, we recorded our level of confidence (high, medium-high, medium, or low) that we accurately relocated plots based on how well our plot matched the original location coordinates and plot descriptions. Despite some relocation uncertainty, the majority (80%) of plots were confidently relocated (rated High or Medium-High).

Plot resurveys

When resurveying plots, we followed the protocol for all BEC plot surveys outlined in the *Field Manual for Describing Terrestrial Ecosystems* (2nd Edition), published by the BC Ministry of Environment and Forests (British Columbia Ministry of Forests & British Columbia Ministry of Environment, 2010). At each plot, a 20.0 × 20.0 m plot was established along the slope aspect, with four 10.0 × 10.0 m quadrats defined within each plot (Figure S4). The full plot was then traversed to identify all herb, shrub, and tree species present. Although not used in this analysis, for each species, percent cover was estimated within each quadrat and averaged across the plot to 1% precision. A full detailed description of field surveying protocols is provided in the Supplementary section S4.

Surveying methods for relocation error plots

Plots surveyed originally in 2019 were selected for the relocation error analysis as they had high GPS coordinate accuracy and were all completed by the same surveyor (H. Klassen) which reduced uncertainty. Field resurveying methods for 2019 plots followed the protocol described above, except that percent cover was only estimated in two diagonally positioned quadrats rather than all four, to reduce sampling time. Species observed in the unsampled quadrats were assigned a cover value of 0.1% in the field. Cover estimates from the two sampled quadrats were averaged to represent the full plot.

To estimate relocation error, during the 2025 resurveys, both a $2025_{Correct}$ plot and a 2025_{Error} plot were established at each of the nine 2019 plots. For the $2025_{Correct}$ plots, we relocated the exact plot center of the 2019 plots, using our relocation protocol, and resurveyed them. These plots were, on average, 6.3 m from the original 2019 coordinates. For the 2025_{Error} plots, we selected the furthest possible location from the estimated true 2019 plot center that maintained similar percent slope gradient, slope aspect, elevation, and overall tree canopy composition, based on the original plot descriptions, and resurveyed this location using the same sampling protocol. This approach was used to simulate realistic relocation uncertainty, where a surveyor may misidentify the exact plot location while still matching key environmental and plot characteristics. The 2025_{Error} plots were therefore designed to represent a worst-case, yet plausible, relocation error

scenario under typical field conditions. On average, the $2025_{Correct}$ and 2025_{Error} plots were separated by a distance of 59.4 m (median = 66.1 m, SD = 12.2 m, range = 40.2–71.2 m).

Statistical analyses

All statistical analyses were conducted in R version 4.6.0 (R Core Team, 2026).

Detecting upslope range shifts

We used Bayesian generalized linear mixed-effects models to quantify changes in species occurrence (present vs. absent) across elevation between historical (1976–2000) and present (2025) surveys. The model was fitted using the brm() function in the brms package (Bürkner, 2017, 2018), with a Bernoulli distribution and logit link. Model structure for random effects was determined through model selection conducted previously using these data (Lo, 2026), where candidate models were compared using Akaike's Information Criterion (AIC). AIC did not support the inclusion of terms for different surveyors or longitude. The selected model included population level effects of time (categorical- historic vs. present), elevation (linear and quadratic), latitude, and interactions between time and elevation (linear and quadratic). We also included species-specific random slopes for elevation (linear and quadratic) and a random intercept for plot ID:

$$\text{Occurrence} \sim \text{Time} \times (\text{Elevation} + \text{Elevation}^2) + \text{Latitude} + (\text{Elevation} + \text{Elevation}^2 \mid \text{Species}) + (1 \mid \text{Plot ID})$$

Elevation and latitude were centred and scaled prior to analysis. Models were fitted using Markov chain Monte Carlo (MCMC) sampling with the No-U-Turn Sampler (NUTS), an adaptive form of Hamiltonian Monte Carlo (HMC), implemented in Stan through brms. Each model was run using four chains, with 4,000 iterations per chain, including 1,000 warm-up iterations. We first fitted the model using the default priors specified by brms and then refitted it using relocation-error-informed priors (*see Incorporating relocation error*).

Incorporating relocation error

Rather than treating relocation error as a separate analysis, we incorporated it directly into the model above, as prior distributions, using the paired $2025_{Correct}$ and 2025_{Error} plots to quantify the variation in species occurrence associated with relocation error. To estimate the error-informed priors, a separate Bayesian mixed-effects model was fitted with the same model structure as above, but with the temporal contrast replaced by a comparison between "correct" vs "error" plots.

The posterior estimates from this relocation-error model provided an empirical estimate of the magnitude and uncertainty of changes attributable to relocation error. We calculated the mean and standard deviation of these posterior distributions and manually set relocation-error-informed normal priors for time, time × elevation, and time × elevation² effects in the full model using the 'set_prior' function in brms. This allowed uncertainty associated with

plot relocation to be incorporated directly into posterior distributions of temporal change, rather than assuming that observed changes between historical and present surveys were entirely attributable to ecological change (*see Supplementary S5 for the informative priors used and S8 for model diagnostics*). As a sensitivity analysis to determine whether estimates of elevational change were sensitive to the relocation-error-informed priors, we compared posterior distributions and predicted range shifts between the error-informed model and a model fitted without relocation-error-informed priors (i.e., with default priors; *see Supplementary S7*).

Model convergence was evaluated using Rhat (all values = 1.0), effective sample sizes (ESS > 1,000), and divergent transitions (none detected; *see Supplementary S6*). Model fit was assessed using posterior predictive checks (function 'pp_check' in brms) and Bayesian R^2 values (function 'bayes_R2' in brms). We also compared prior and posterior distributions to assess the extent to which the observed data updated the relocation-error-informed prior distributions using the 'sample_prior=only' argument in the brm function of brms. Model posteriors were summarized (brms 'summary' function) using posterior means, standard errors, and 95% credible intervals (CrIs). We considered a parameter to have evidence for an effect when its 95% CrI excluded zero.

To quantify range shifts, we generated posterior predictions across the elevational gradient for each time period using the 'posterior_epred' function in brms and identified the elevation of the maximum predicted occurrence separately for each posterior draw in both time periods. The range shift was calculated as the difference between the present and historical predicted optima, summarized using the posterior medians, standard deviations, and 95% CrIs, with positive values indicating an upslope shift.

Functional group analysis

We extended the Bayesian framework above to determine whether elevational shifts differed among plant functional groups (herb, shrub, or tree) by adding a functional group interaction to the main model:

$$\text{Occurrence} \sim \text{Time} \times (\text{Elevation} + \text{Elevation}^2) \times \text{Functional group} + \text{Latitude} + (\text{Elevation} + \text{Elevation}^2 \mid \text{Species}) + (1 \mid \text{Plot Number})$$

A separate relocation-error model was fitted using the paired 2025_{Correct} and 2025_{Error} plots as described above and posterior estimates from this model were used to construct relocation-error-informed priors for the corresponding functional group interactions model (*see Supplementary S9)*. The functional group model was then fitted using the same Bayesian framework and model diagnostics as the main model (*see Supplementary S10*). Posterior predictions were obtained using posterior_epred() and used to estimate elevational optima and range shifts separately for herbs, shrubs, and trees. Additionally, to assess whether each functional group differed in their probability of occurring between historical and present surveys, differences in the peak predicted occurrence probability were calculated for each posterior draw.

This was done by subtracting the historical maximum of predicted probability from the present maximum for each functional group. Posterior medians, standard deviations, and 95% CrIs were calculated for these differences, along with the posterior probability of an increase, estimated as the proportion of draws with a positive difference.

Results

Changes in species occurrence across elevation while accounting for relocation error

Consistent with our hypotheses, species exhibited an upward shift in their elevational distributions between historical (1976–2000) and present surveys (2025), while accounting for relocation error (Figure 2). The posterior distribution for the interaction between time and linear elevation was positive with a mean of 0.36 and a CrI excluding 0 (SE = 0.15, 95% CrI = 0.06–0.65; Table 1), providing evidence that the probability of species occurrence increased at higher elevations in the present relative to the historical period. Notably, while the predicted probability of species' occurrence in the present tended to be greater than the historical probability of occurrence across all elevations, the difference was greatest at mid-to-high elevations (Figure 2). Range shifts estimated from the posterior distribution were positive, with a median shift of +55 m (SD = 31 m, 95% CrI = +7 to +127 m; Figure 2; Table 2).

To assess how the relocation-error-informed priors contributed to these estimates, we compared the prior and posterior distributions. Posterior distributions were substantially narrower than their corresponding relocation error priors, indicating that the observed data substantially updated the prior distributions (Figure S6.A). The model explained approximately 39% of the variation in species occurrence (Bayesian $R^2 = 0.39$, 95% CrI = 0.36–0.41).

Lastly, to assess the influence of the relocation-error-informed priors on model estimates, we compared the fit of the full model to the model fit without these informative priors as a sensitivity analysis. Interestingly, the results were nearly identical to those of the model incorporating relocation error, with a time × elevation posterior mean of 0.36 (*see Supplementary section S7;* SE = 0.15, 95% CrI = 0.07–0.65; Table S7.A) and an estimated range shift of +56 m (SD = 31 m, 95% CrI = +9 to +128 m; Table S7.B). These results indicate that the estimated upslope shifts were highly robust to the relocation-error-informed priors.

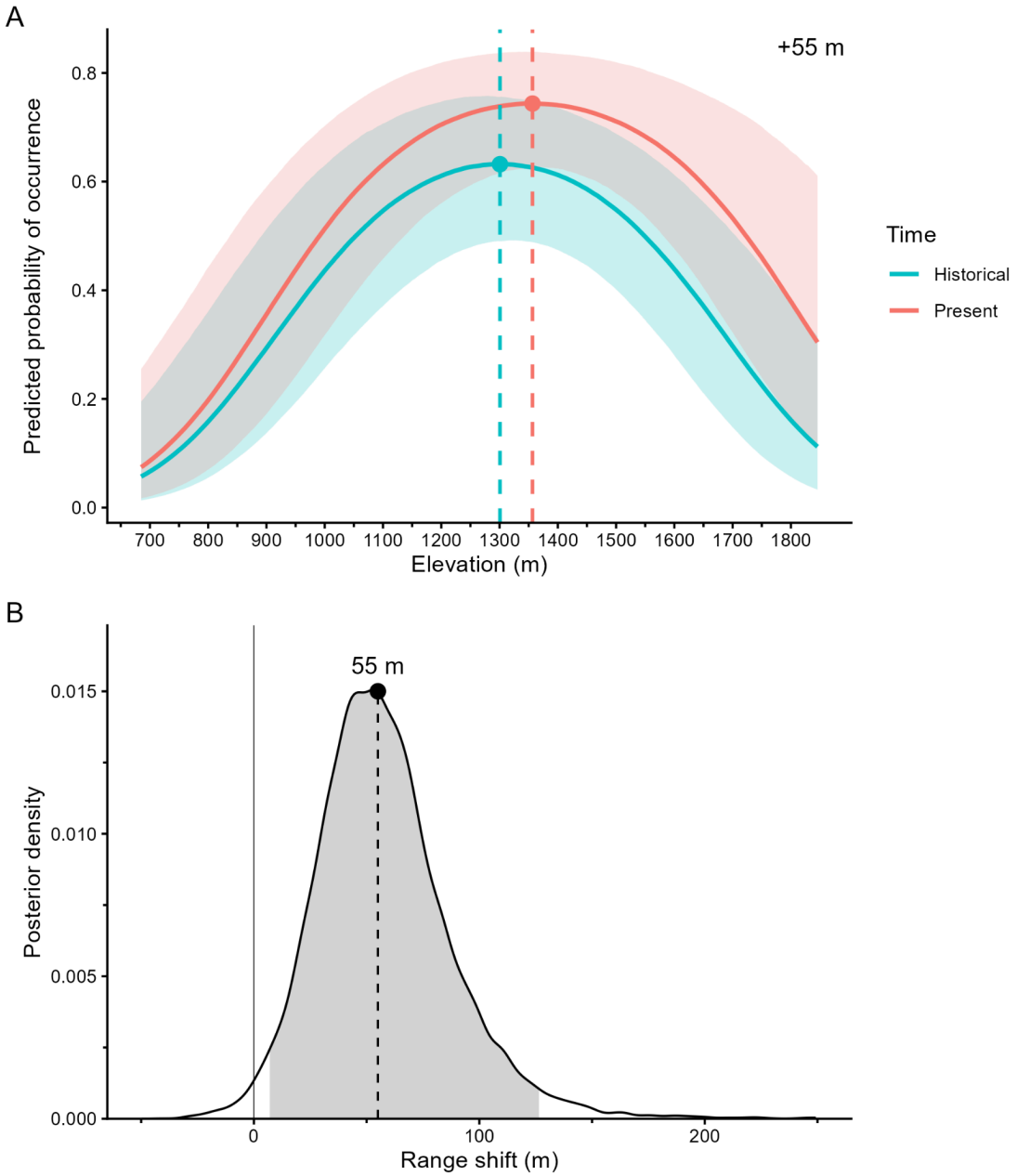


Figure 2. Predicted elevational shift in species occurrence between historical and present surveys. (A) Predicted probability of species occurrence across elevation for the historical (1976–2000, blue) and present (2025, red) periods, with shaded areas representing 95% credible intervals. Dashed vertical lines indicate the median elevation of peak predicted occurrence for each period (B) Posterior distribution of estimated range shifts, with the dashed vertical line indicating the median shift and the grey shaded region indicating the 95% credible interval.

Table 1. Posterior estimates for the full Bayesian model. Posterior means, standard errors, and 95% credible intervals are shown for each regression coefficient. Bolded credible intervals exclude 0.

Parameter	Estimate	Est. Error	95% CrI
Intercept	0.53	0.29	−0.04, 1.11
Time: Present	0.53	0.16	**0.22, 0.85**
Elevation (linear)	−0.25	0.38	−0.97, 0.50
Elevation²	−1.26	0.22	**−1.70, −0.84**
Latitude	0.77	0.21	**0.38, 1.21**
Time × Elevation	0.36	0.15	**0.06, 0.65**
Time × Elevation²	0.13	0.14	−0.15, 0.40

Table 2. Predicted elevational optima and range shift between historical (1976–2000) and present (2025) surveys. Estimates represent posterior medians, with posterior standard deviations (SDs) and 95% credible intervals (CrIs) reported for each optimum and the estimated range shift.

Measure	Posterior median (m) ± SD (m)	95% CrI (m)
Historical optimum	1301 ± 58	1196–1424
Present optimum	1357 ± 69	1238–1511
Range shift	+55 ± 31	+7 to +127

Functional group differences in elevational change

The Bayesian mixed-effects model revealed differences in elevational responses among functional groups (Figure 4; Table 3; Table 4). Herbaceous species showed the clearest evidence of an upslope range shift, with a median shift of +135 m (SD = 105 m; 95% CrI = +19 to +441 m; Table 3). Their peak predicted probability of occurrence also tended to increase between historical and present surveys, and was most pronounced at mid-to-high elevations (Figure 4), with a median difference of +0.12 (SD = 0.08; 95% CrI = −0.01 to +0.29; Table 4). However, the 95% CrI marginally overlapped zero, despite a high posterior probability of an increase (0.964; Table 4).

Shrub species showed an increase in peak predicted occurrence probability between historical and present surveys, with a median difference of +0.15 (SD = 0.05; 95% CrI = +0.05 to +0.25; Table 4) and a 0.998 posterior probability of an increase (Table 4). However, they showed little evidence of an upslope range shift (median range shift = +9 m, SD = 44 m; 95% CrI = −71 to +100 m; Table 3).

Lastly, tree species showed little evidence of an upslope range shift (median range shift = +56 m, SD = 127 m; 95% CrI = −200 to +364 m; Table 3) or a change in peak predicted occurrence probability (median difference = +0.01, SD = 0.07; 95% CrI = −0.12 to +0.16; Table 4).

As with the previous model, diagnostics indicated good convergence, sampling performance, and overall model fit, while comparison of the prior and posterior distributions indicated the prior distribution was sufficiently updated by the data (*see Supplementary S10*).

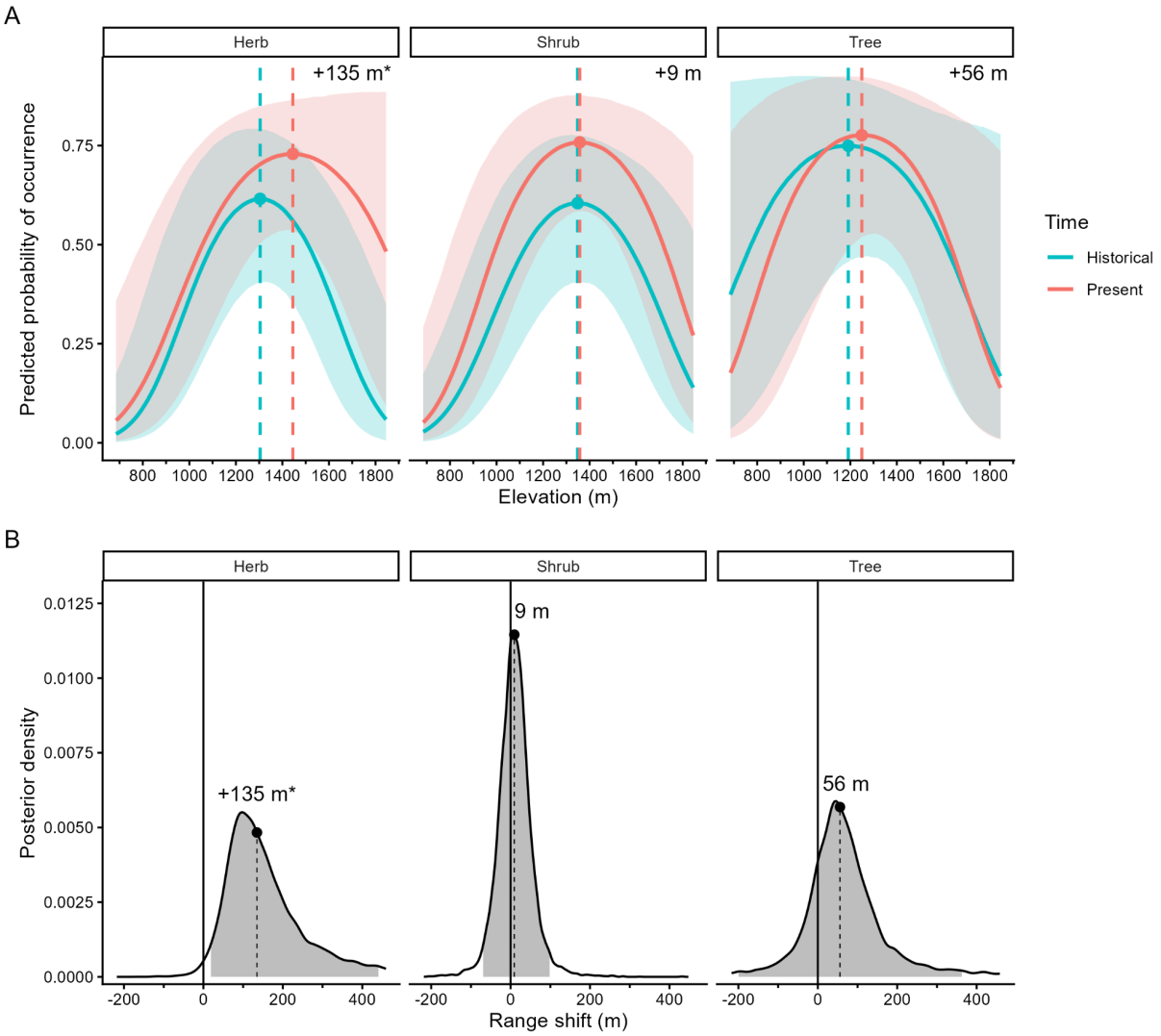


Figure 3. Predicted elevational shifts in species occurrence among functional groups. (A) Predicted probability of occurrence across elevation for historical (1976–2000, blue) and current (2025, red) surveys for herb, shrub, and tree species. Shaded areas represent 95% credible intervals, and dashed vertical lines indicate the median elevation of the peak predicted occurrence. (B) Posterior distributions of estimated range shifts for each functional group, with dashed vertical lines indicating median range shifts and grey shaded regions representing 95% credible intervals. The * indicates a 95% credible interval that excludes zero.

Table 3. Predicted elevational optima and range shifts for herb, shrub, and tree species between historical (1976–2000) and present (2025) surveys. Values represent posterior medians, with standard deviations (SDs) and 95% credible intervals (CrIs) reported for each optimum and estimated range shift. Positive range shifts with CrIs excluding zero indicate upslope shifts in peak occurrence.

Measure	Herb (m)	Shrub (m)	Tree (m)
Historical optimum (median ± SD)	1304 ± 76	1348 ± 94	1192 ± 216
Historical 95% CrI	1168–1472	1200–1567	685–1771
Present optimum (median ± SD)	1445 ± 156	1358 ± 96	1250 ± 146
Present 95% CrI	1215–1846	1205–1587	982–1581
Range shift (median ± SD)	+135 ± 105	+9 ± 44	+56 ± 127
Range shift 95% CrI	+19 to +441	−71 to +100	−200 to +364

Table 4. Peak predicted occurrence probabilities and differences between historical (1976–2000) and present (2025) surveys for herb, shrub, and tree species. Estimates represent posterior medians, with posterior standard deviations (SDs) and 95% credible intervals (CrIs) reported for probability differences. Positive differences indicate an increase in the peak predicted occurrence probability from historical to present surveys, while negative differences indicate a decrease. The posterior probability of an increase represents the proportion of posterior draws in which the difference was positive.

Functional group	Historical peak probability (median ± SD)	Present peak probability (median ± SD)	Difference (median ± SD)	95% CrI for difference	Posterior probability of increase
Herbs	0.63 ± 0.10	0.75 ± 0.09	0.12 ± 0.08	−0.01–0.29	0.964
Shrubs	0.62 ± 0.09	0.77 ± 0.07	0.15 ± 0.05	0.05–0.25	0.998
Trees	0.78 ± 0.11	0.80 ± 0.10	0.01 ± 0.07	−0.12–0.16	0.592

Discussion

This study provides evidence of widespread elevational shifts of montane plant species that are robust to relocation error. On average, plant species shifted upwards by 55 m between historical (1976–2000) and present (2025) surveys. However, this average was driven almost entirely by herbaceous species. Herbs showed evidence of both an upslope shift by 135 m and increase in occurrence concentrated at mid-to-high elevations, while shrubs showed no upslope shift but a clear increase in occurrence overall, and trees showed no directional change in their range or occurrence through time. This study provides additional empirical support for the expected and most commonly observed trend of upward shifts with climate warming (Lawlor et al., 2024) while also providing a generalizable framework for explicitly incorporating relocation error into estimates of ecological change.

Robustness of range shifts to relocation error

A primary challenge in detecting species range shifts from historical resurveys is addressing sources of bias, their extent, and whether or not observational studies can capture true temporal change. Relocation error is particularly important to consider because spatial mismatch between historical and contemporary plots can generate pseudoturnover as an artifact of sampling error and spatial heterogeneity (Fischer & Stöcklin, 1997; Kapfer et al., 2016). Such concerns have contributed to broader uncertainty about the reliability of using vegetation resurveys to study the impacts of modern climate change on species diversity and distributions. In our study, however, model estimates of elevational change through time with realistic relocation error incorporated, were not only credible (Table 1), but nearly identical to those estimated from the model with no relocation error (Table S7.A). This does not, however, suggest that relocation error is unimportant in resurvey studies generally, but rather that, with sufficient data coverage, the likelihood provides enough evidence that the posterior estimates of change are driven primarily by the observed data than by the relocation error-informed prior distribution. In other words, we captured sufficient information about changes in our system that the error prior became negligible; determining at what level of information this becomes true is a challenge ideally suited to Bayesian inference.

A key mechanism underlying the robustness to relocation error observed in our study is likely the relatively large 400 m² (20 × 20 m) plots, which may capture sufficiently representative community composition to buffer against moderate spatial mismatches. This is consistent with Kopecký and Macek (2015), who found that resurveys can be robust to relocation error and proposed that larger plots reduce relocation-driven pseudoturnover. In contrast, Boch et al. (2019) found species turnover increased significantly with relocation error in much smaller 28 m² plots, which may have been more sensitive to fine-scale spatial heterogeneity. Hence, relocation error may be scale dependent, with smaller plots more prone to pseudoturnover and larger plots potentially buffering these effects.

Plot size alone, however, is unlikely to explain the limited effect of relocation error observed here, as it likely depends on several interacting factors. Verheyen et al. (2018) emphasized that relocation error also depends on the quality of historical location data and survey effort, while Shih Mendez et al. (2026) found that the effects of plot relocation strategy varied among vegetation types and ecological metrics. In this study, the use of GPS coordinates, detailed plot descriptions, and standardized BEC survey protocols during relocation and resurveying likely reduced relocation error and observer-related variability. Additionally, the historical BEC plots were selected within relatively homogeneous landscape units to maximize their representation of broader ecosystem types (Pojar et al., 1987). The limited effect of relocation error observed here may therefore reflect the combined effects of plot size, landscape homogeneity, and the quality and consistency of the historical BEC data, which together may have reduced sensitivity to moderate spatial mismatches during relocation.

The framework used here therefore provides a way for future resurvey studies to evaluate the potential influence of relocation error within their own study systems. Rather than assuming that relocation error either does or does not bias estimates of ecological change, researchers could empirically quantify variation associated with relocation error and determine whether incorporating that uncertainty changes inferences. In some systems, as in E.C. Manning Park, the effect may be negligible relative to the ecological signal; in others, particularly where plots are small, vegetation is highly heterogeneous, or historical locations are poorly constrained, relocation error may have a greater influence.

Functional group differences in elevational responses

The importance of distinguishing methodological uncertainty from biological variation becomes particularly clear when considering the substantial variability in documented range shifts globally. For example, Lawler et al. (2024) found that while 59% of documented range shifts are directionally consistent with warming, 35% are in the opposite direction of expectations, and 6% found no shift at all. Methodological biases alone are unlikely to explain the substantial variation in observed range shifts. Species also differ fundamentally in their ability to respond to changing environmental conditions, including life history, functional traits, and overall tolerance (Collins et al., 2025; Henn et al., 2024; Svenning & Sandel, 2013). In our study, the contrasting responses amongst growth forms provide an example of the life history component of this variation.

Here, herbs showed the clearest evidence of an upslope range shift, with a median shift of 135 m (Table 3), suggesting that species with faster life histories may be tracking climate more quickly and effectively. This response is consistent with evidence that shorter-lived, faster-turnover taxa can track changing climatic conditions more rapidly than longer-lived woody species, which may lag behind shifting climatically suitable habitat due to slower reproduction and mortality (Henn et al., 2024; Lenoir et al., 2008). In general, herbs tend to have shorter generation times and faster demographic turnover than woody species, potentially facilitating more rapid responses to changing conditions (Jump & Peñuelas, 2005). Moreover, herbs tended to show an increase in predicted occurrence that was greatest at mid-to-high elevations (Figure 3). To date, one of the strongest signals of changing range dynamics is an increase of species richness on mountain summits (Pauli et al., 2012; Steinbauer et al., 2018; Wessely et al., 2026). Although this study does not directly test species richness, an increased probability of occurrence at higher elevations is consistent with patterns of increasing richness in other montane systems.

Contrary to herbs, shrubs did not show evidence of an upslope shift (Table 3). However, their predicted probability of occurrence did increase substantially between historical and present surveys (Figure 3; Table 4), suggesting that shrub species may be increasing in prevalence within their existing elevational distributions. In high latitude and elevation ecosystems, similar increases in shrub cover and abundance have been widely documented and are often attributed to warming-driven shrub encroachment, otherwise known as shrubification (Myers-Smith et al., 2011; Vanneste et al., 2026). Shrubification has been observed in subalpine ecosystems under

warming and changes to disturbance regimes (Zeidler & Banaš, 2024), and is increasingly recognized as an important signal of climate change. While it has been shown to have positive effects by maximizing plant diversity under moderate abundances (Eldridge & Soliveres, 2015), it has also demonstrated negative effects by altering community composition, reducing herbaceous diversity and population growth rates (Collins et al., 2019; Myers-Smith et al., 2011). In the context of this study, the increase in shrub occurrence may reflect in-filling of species across plots or that shrubs may be responding to environmental change differently than herbaceous species. The increased occurrence of shrubs may be driven by increased shrub richness or cover, and a future direction would be to directly assess changes in shrub abundance and richness across plots to determine whether this pattern reflects an influx of new species or an increased detection of existing species.

Unlike herbs and shrubs in our study, which have displayed upslope range shifts and increased prevalence, the lack of detectable change by trees would suggest these long-lived woody species are lagging behind shifting climate conditions (Table 3; Table 4). Trees are often constrained by long generation times, slow or episodic recruitment compared to shorter-lived species, which can delay their ability to track changing environments at quick enough timescales (Aitken et al., 2008). As a result, it has been widely documented that tree species may be persisting in climatically suboptimal environments, leading to a state of climatic disequilibrium, where current distributions are underfilling their full climatically suitable ranges (Seliger et al., 2021). At the same time, the absence of a detectable change in tree occurrence does not necessarily indicate demographic stasis. Changes in recruitment, growth, or stand age/size composition may not be detectable from presence–absence models, as early signals of range shifts may be more apparent in the establishment of young individuals rather than changes in overall occurrence (Bohner & Diez, 2020). Tree height data and information in the recruitment and seedling layers of the BEC database could therefore provide an opportunity to test whether changes in recruitment or stand structure are occurring despite the lack of detectable range shifts.

Together, these results demonstrate that the upslope shift detected across species does not represent a uniform response of the montane flora. Differences in responses across functional groups are consistent with the idea that species track environmental change through multiple pathways depending on their life history and demographic characteristics. They also provide a potential biological explanation for some of the variation in range shifts documented globally: species can experience similar regional environmental trends while differing substantially in their ability to disperse, establish, persist, and respond.

Conclusions and future directions

Our study provides evidence of an upslope shift in plant species distributions in E.C. Manning Provincial Park that was robust to moderate relocation error, while also demonstrating substantial differences in responses amongst growth forms. These findings highlight the value of historical

vegetation resurveys for detecting long-term ecological change, but also the importance of carefully accounting for uncertainty when interpreting patterns of change. In this study, we strengthened ecological inference by reducing methodological uncertainty through the use of multiple sources of information for plot relocation, standardized survey protocols, trained observers, and restricting analyses to the most frequently occurring species, while explicitly quantifying relocation error and incorporating the uncertainty from pseudoturnover into our models. While previous studies have recommended approaches for minimizing sources of error in vegetation resurveys (Kapfer et al., 2016; Shih Mendez et al., 2026; Verheyen et al., 2018), our framework provides a way to explicitly account for residual relocation error in estimates of ecological change. Rather than discouraging the use of historical vegetation records, these considerations can strengthen the inferences drawn from them. The BC Ministry of Forests Biogeoclimatic Ecosystem Classification (BEC) database provides an especially valuable opportunity in this regard, containing more than 70,000 vegetation plots distributed across approximately 11° of latitude and numerous mountainous regions throughout British Columbia, Canada. Extending this approach to contemporary resurveys of additional BEC plots could provide an unprecedented opportunity to assess long-term vegetation change across broad environmental gradients. More broadly, integrating rigorous approaches to addressing methodological uncertainty into future studies will support the continued use of vegetation resurveys for monitoring long-term ecological change.

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Supplementary

S1. Estimated warming averaged across plots in E.C. Manning Provincial Park

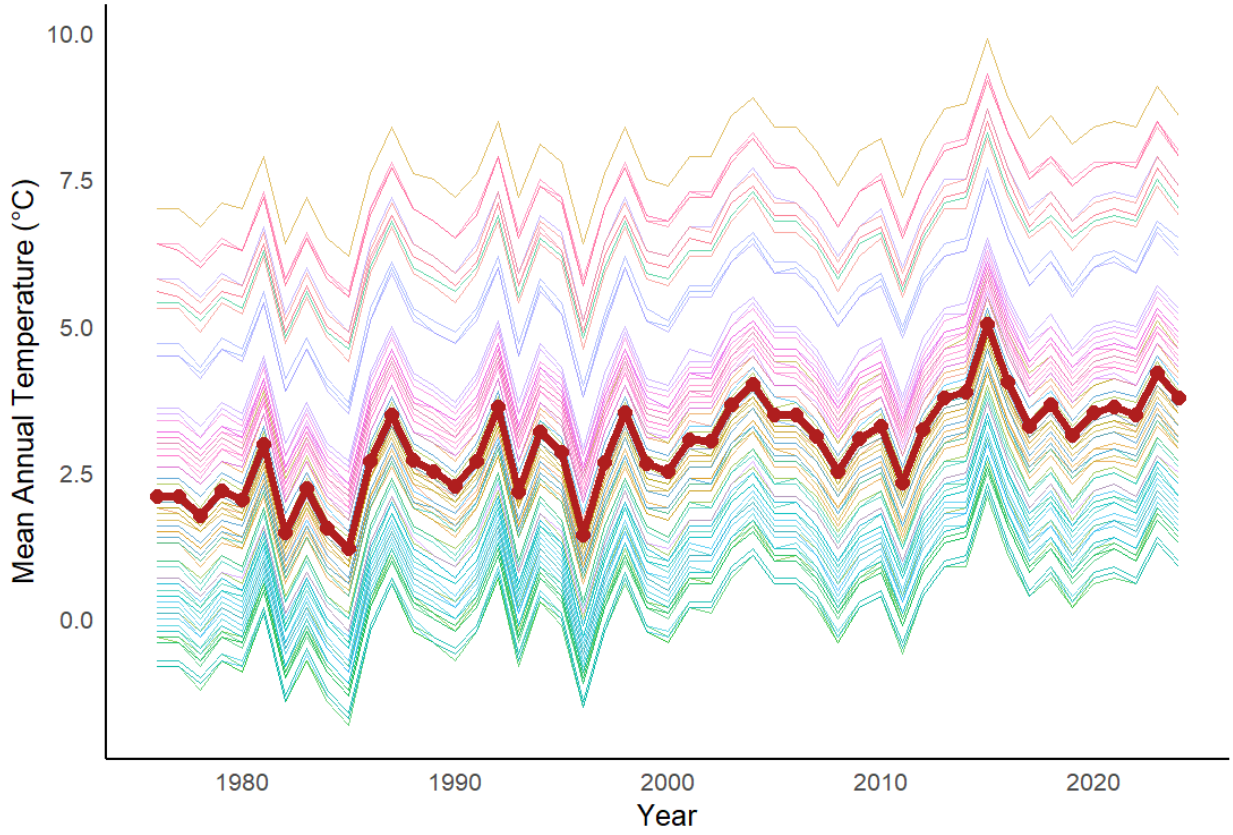


Figure S1. Mean annual temperature (MAT °C) in E.C. Manning Provincial Park from 1976 to 2025. Thin coloured lines represent annual MAT at individual locations across the park, and the thick red line represents the mean across locations for each year. The estimated warming of 1.53 °C from 1976 to 2025 was calculated using a linear regression of mean annual temperature against year.

S2. Correlation of elevation vs. sampling year.

To address the correlation between elevation and sampling year, historical plots from 1976-2000 were binned together as the historical baseline. Plots surveyed after the year 2000 were not used in the analysis, with the exception of plots from 2019, used in the relocation analysis.

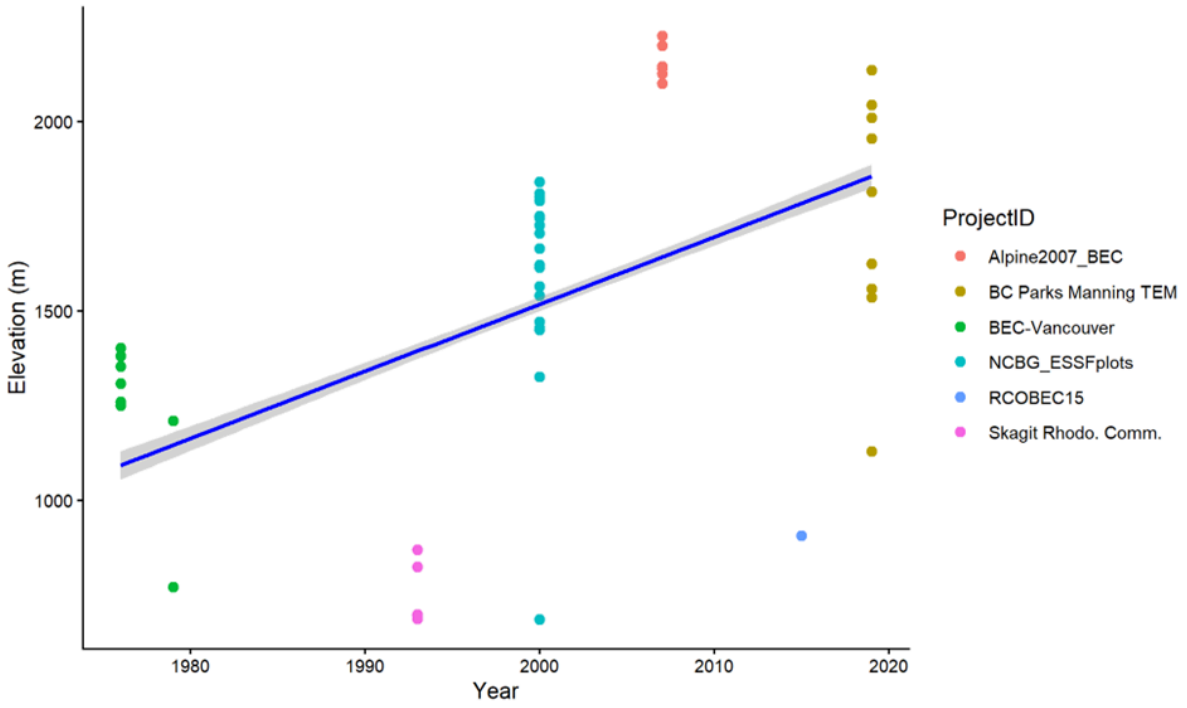


Figure S2. The relationship between BEC sampling year and elevation (cor = 0.511, p = 2.2e-16). Each colour corresponds to a different project, and each dot represents a plot resurveyed within that project.

S3. Evaluation of species richness differences between datasets

Species richness was calculated at the plot level for BEC and 2025 datasets (excluding grasses, mosses, and lichens). Differences were assessed within each project using Wilcoxon rank-sum tests. Across most projects, species richness did not differ significantly between datasets, and richness distributions showed substantial overlap (Table S3.A; Table S3.B). Sampling effort and plot sizes were consistent between datasets, and variability within projects was generally greater than differences between datasets. As a result, rarefaction was not applied, and all analyses were conducted using observed (non-rarefied) data.

Table S3.A. Comparison of species richness between BEC and 2025 datasets by project (excluding grasses, mosses, and lichens). Species richness (SR), year of original survey, standard deviation (SD), and number of plots (N) are represented for each project.

BEC Project ID	Year of original survey	Mean SR (2025)	Mean SR (BEC)	SD (2025)	SD (BEC)	N of plots
BC Parks Manning TEM	2019	25.00	23.10	7.64	5.67	9
BEC-Vancouver	1976 & 1979	37.11	32.56	6.41	4.82	9
NCBG_ESSF plots	2000	26.33	14.17	8.09	4.87	18
Skagit Rhodo. Comm.	1993	24.29	24.00	2.81	4.80	7

Table S3.B. Results of Wilcoxon rank-sum tests comparing species richness between 2025 resurveys and historical BEC plots for each project (excluding grasses, mosses, and lichens). Test statistics (W) and associated p-values are reported for each comparison.

Project ID	W statistic	p-value
BC Parks Manning TEM	58.5	0.544
BEC-Vancouver	60.5	0.084
NCBG_ESSFplots	292	<0.001
Skagit Rhodo. Comm.	24.5	0.402

S4. Full field surveying methods description

Field surveying methods

All surveying methods followed the protocol outlined in the BC Ministry of Environment and Forests' Field Manual for Describing Terrestrial Ecosystems (2nd Edition), which has remained unchanged between the 1990 and 2010 publications (British Columbia Ministry of

Forests & British Columbia Ministry of Environment, 2010; Luttmerding et al., 1990). To ensure the protocol was interpreted correctly, we interviewed Provincial BEC Ecologist, Heather Klassen, who surveyed the historical plots from 2015-2019 and confirmed our methods. Additionally, Courtney Collins, a research ecologist with the BC Ministry of Forests, assisted with field surveys at several plots to verify that our techniques aligned with the BEC field protocol. Although vegetation cover data were collected during field surveys as part of the standardized BEC protocol, these data were not used in the analyses presented in this study. We describe the collection of these data below to provide a complete account of field methods and to facilitate comparison with historical surveys.

Once plot centers were confidently relocated, 20.0 m × 20.0 m plots were set up along the slope aspect using a compass and an electronic sonar distance-measuring device (Sonin Combo Pro 10300). From the plot center, we measured 10.0 meters upslope and placed flags at 10.0 m intervals around the 20.0 m × 20.0 m plot, creating four quadrats within each plot (Figure S4). Site features were recorded using a clinometer and compass, and detailed plot descriptions were documented, including location, community composition, plant condition, disturbances, and any relocation reasoning if plot centers were adjusted.

We began resurveying by traversing the full plot and identifying all tree, shrub, and herb species present. Species were recorded using standardized Latin names and BEC species codes (four-letter genus and three-letter species abbreviations) to be directly comparable to the original survey data and consistent with provincial standards (British Columbia Ministry of Forests & British Columbia Ministry of Environment, 2010). Unknown species were photographed and given a temporary name until they could be identified later. To improve accuracy, we estimated cover for each species within each of the four quadrats and then averaged these values across the full plot. Within each quadrat, percent cover was recorded in 1% intervals for values greater than 1%, while cover between 0-1% was estimated with 0.1% precision.

Grasses, mosses, and lichens were not identified to the species level, and instead, their cover was recorded as functional groups (one for grasses and one for mosses and lichens). Analysis of these functional groups was beyond the scope of this study and therefore not analyzed. Analysis of tree height data was also not included in this analysis; however, tree cover

was estimated by species within three height classes: A2 (>10 m), B1 (2–10 m), and B2 (<2 m). Tree heights were measured using a hypsometer (TruPulse 200X). Tree cover was then derived by adding the cover for each species present in a plot across its height classes. Lastly, although not used in this thesis, total tree canopy cover across all species was estimated within each quadrat and averaged to the plot level.

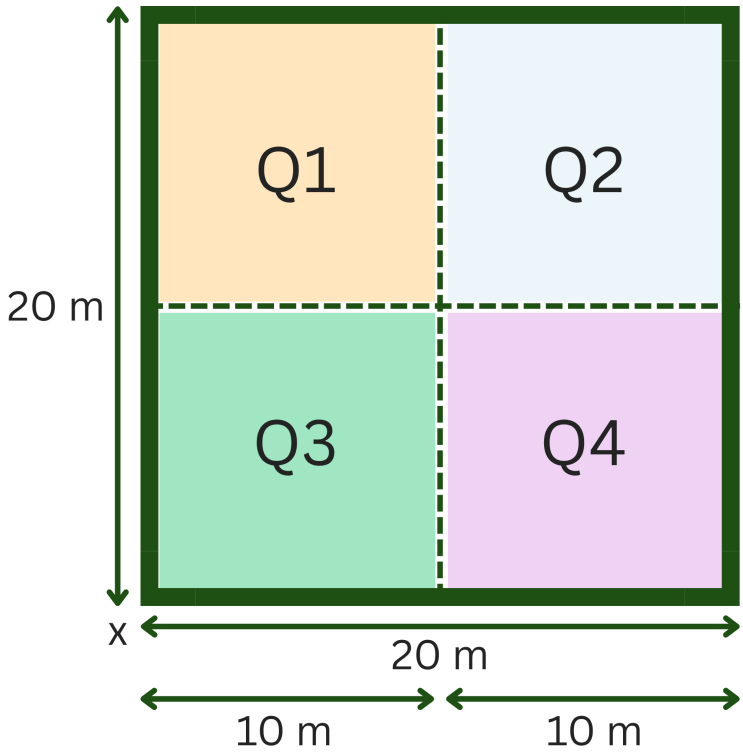


Figure S4. Diagram illustrating the quadrat layout within each 20.0 × 20.0 m (400 m^2) plot. Flags were placed at each quadrat corner in 10.0 m intervals to define boundaries, and percent cover of each species was recorded within each individual quadrat.

S5. Informative priors used in the relocation-error-informed Bayesian model

Table S5. Relocation-error-informed priors used in the full Bayesian model.

Parameter	Prior	μ	SD
Time: Present	Normal	0.72	2.24
Time × Elevation	Normal	0.45	1.41
Time × Elevation²	Normal	−0.18	1.91

S6. Model diagnostics of the full Bayesian model with relocation-error-informed priors

Table S6.A. MCMC convergence and sampling diagnostics for the full Bayesian model. All parameters had Rhat values of 1.00, indicating convergence among the four MCMC chains. Effective sample sizes were high across the fixed effects, with Bulk ESS values ranging from 2,325 to 10,493 and Tail ESS values ranging from 4,063 to 9,067. No divergent transitions were detected during NUTS sampling.

Diagnostic	**Result**
MCMC chains	4
Iterations per chain	4,000
Warmup iterations per chain	1,000
Post-warmup draws	12,000
Rhat	1.00 for all parameters
Bulk ESS range, fixed effects	>1000 for all parameters
Tail ESS range, fixed effects	>1000 for all parameters
Divergent transitions	0

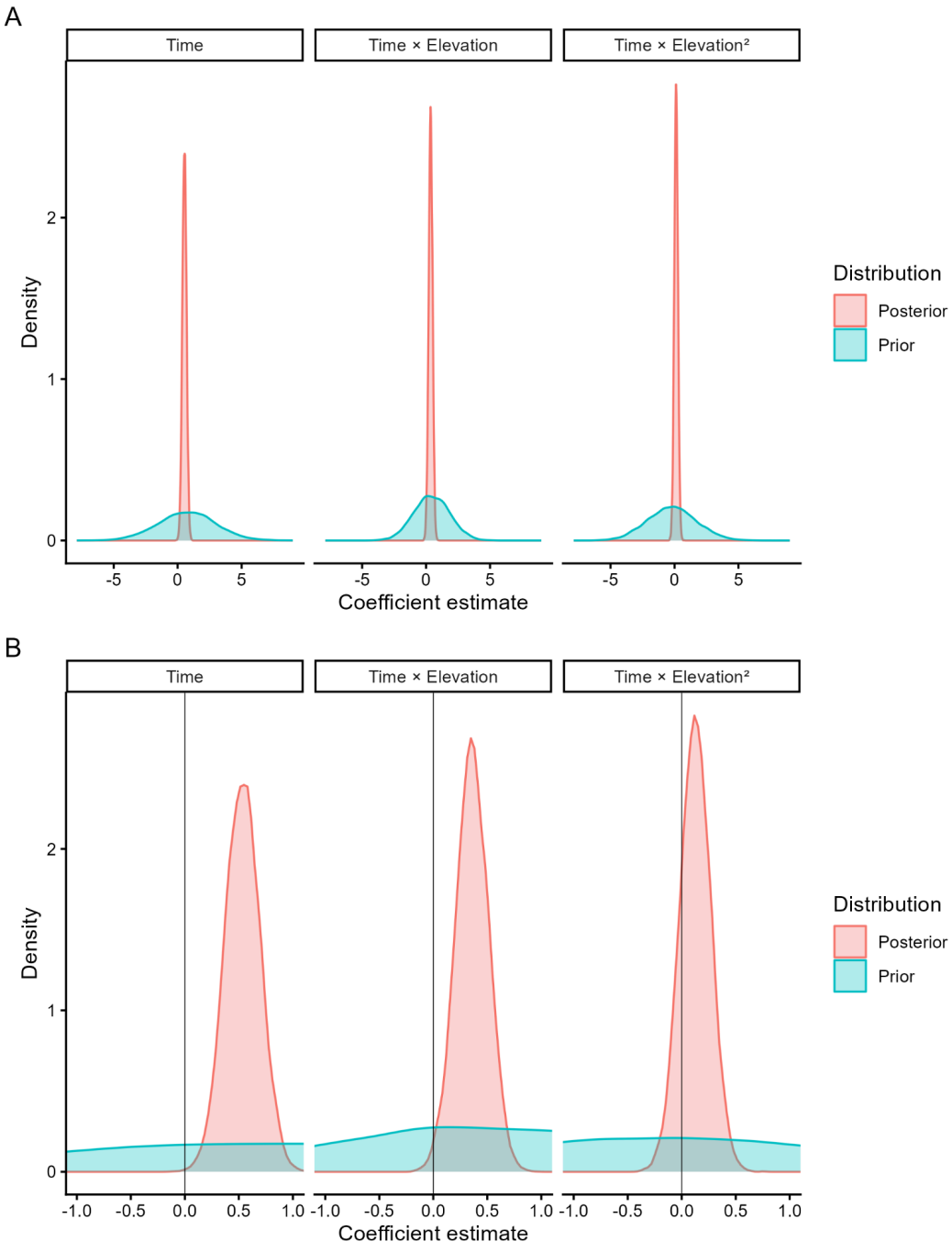


Figure S6.A. Prior (blue) and posterior (red) distributions for key fixed-effect coefficients in the Bayesian model. (A) Relocation-error-informed prior and posterior distributions for the effects of time, time × elevation, and time × elevation². (B) Same distributions shown on a narrower coefficient scale (-1 to 1) to highlight the posterior distributions in proximity to 0.

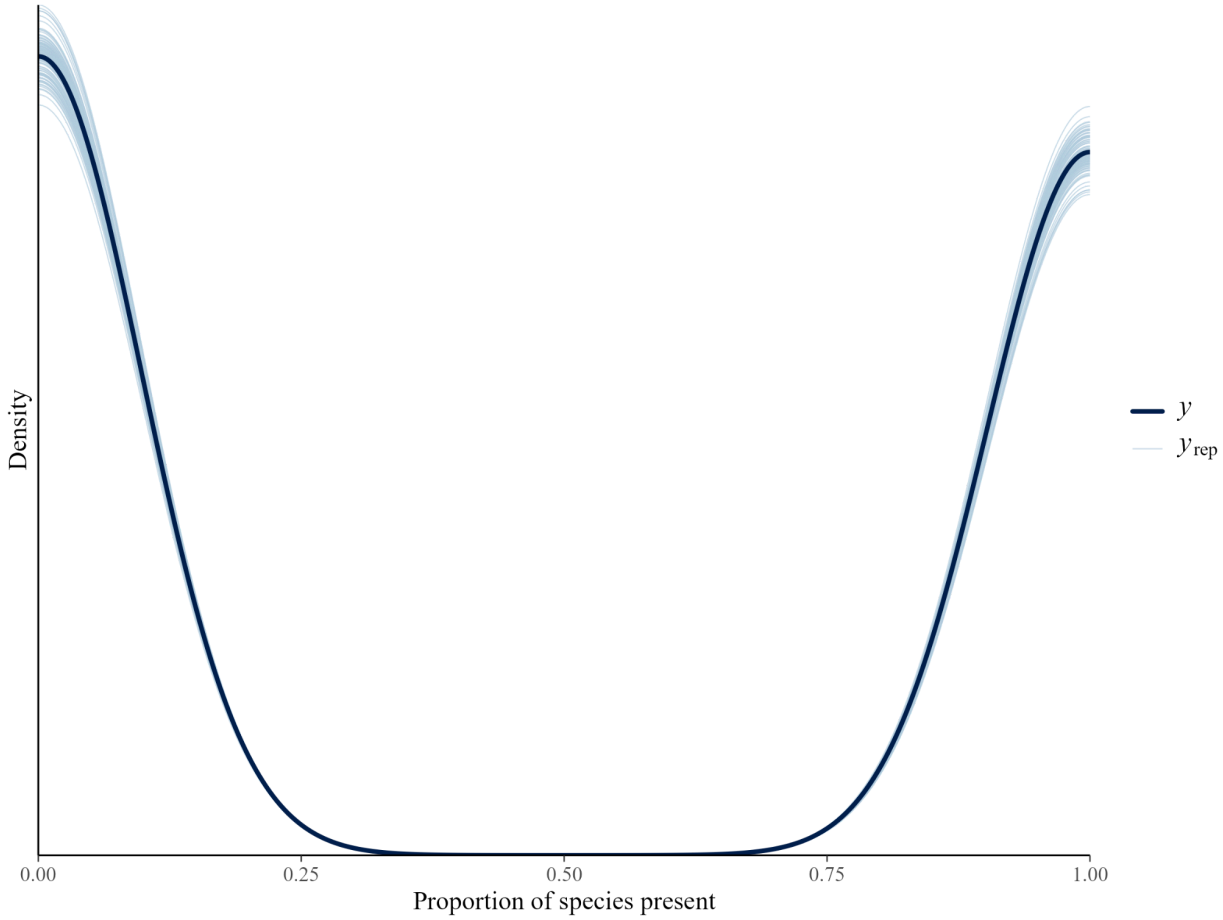


Figure S6.B. Posterior predictive check of the full model with informative priors. The consistency of the real data and simulated data demonstrate that the model accurately characterizes the data in this density graph. The simulated data is shown by the dark blue curve (y) and observed data is shown by the light blue curves (y_{rep}, n = 100).

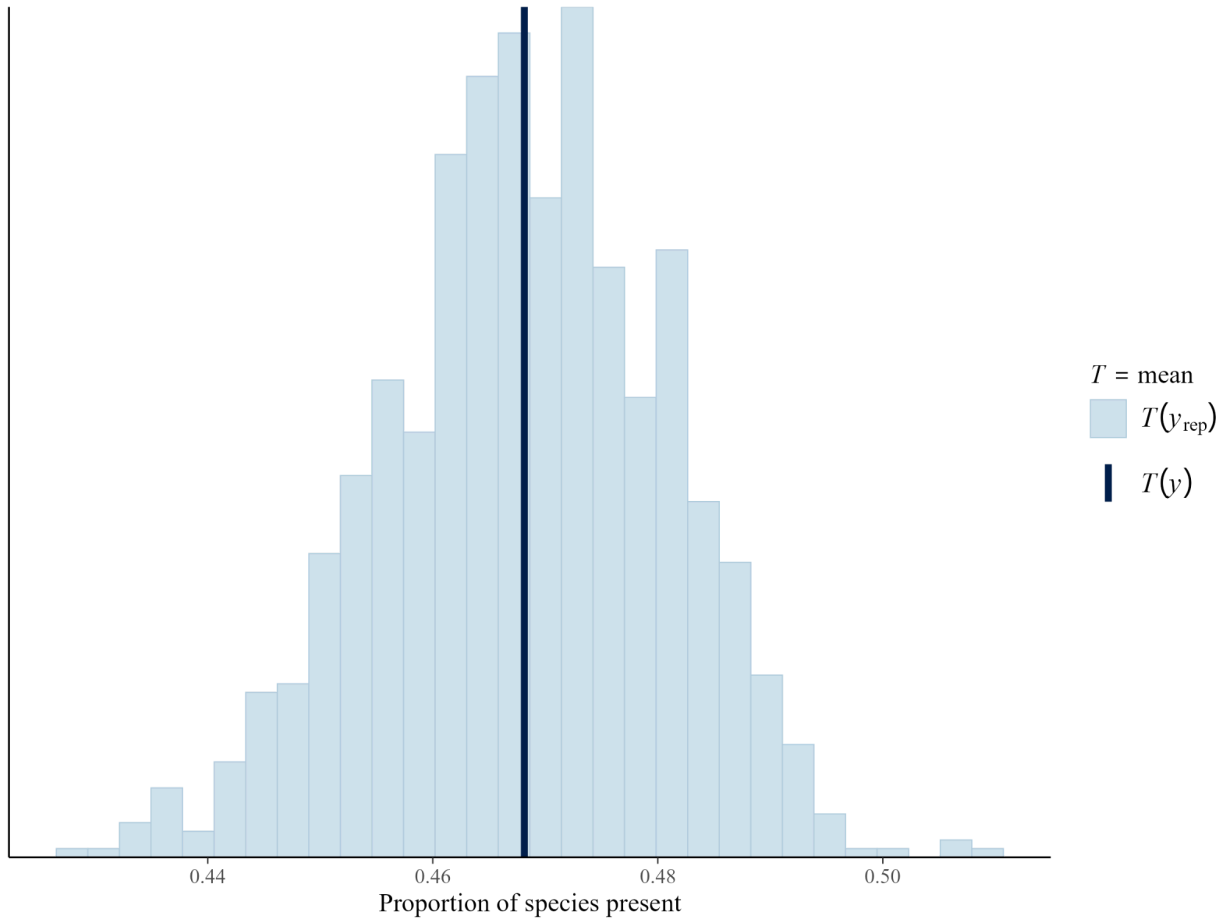


Figure S6.C. Posterior predictive check of the full model with informative priors. The consistency of the real data and simulated data demonstrate that the model accurately characterizes the mean proportion of species present in this histogram. The simulated data is shown by the dark blue line (y) and observed data is shown by the light blue (y_{rep}, n = 1000).

S7. Model diagnostics of the full Bayesian model without relocation-error-informed priors

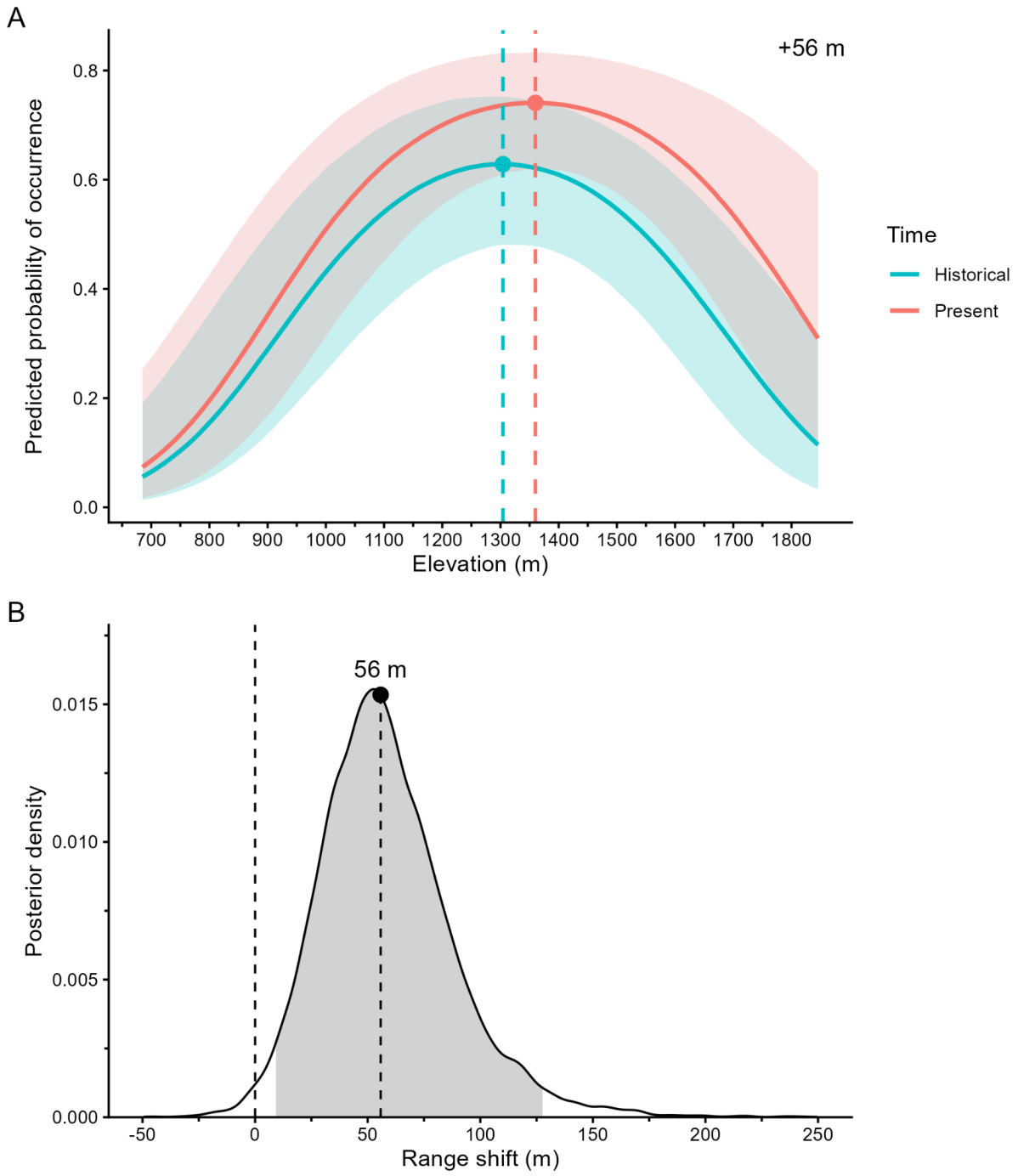


Figure S7.A. Predicted elevational shift in species occurrence between historical and present surveys for the model with no relocation-error-informed priors. (A) Predicted probability of species occurrence across elevation for the historical (1976–2000, blue) and present (2025, red)

periods, with shaded areas representing 95% credible intervals. Dashed vertical lines indicate the median elevation of peak predicted occurrence for each period (B) Posterior distribution of estimated range shifts, with the dashed vertical line indicating the median shift and the grey shaded region indicating the 95% credible interval.

Table S7.A. Posterior estimates for the full model without informative priors. Posterior means, standard errors, and 95% credible intervals are shown for each regression coefficient. Bolded credible intervals exclude 0.

Parameter	**Estimate**	**Est. Error**	**95% CrI**
Intercept	0.51	0.30	−0.08, 1.09
Time: Present	0.53	0.16	**0.22, 0.85**
Elevation (linear)	−0.23	0.37	**−0.97, 0.50**
Elevation²	−1.25	0.22	**−1.68, −0.83**
Latitude	0.77	0.21	**0.38, 1.19**
Time × Elevation	0.36	0.15	**0.07, 0.65**
Time × Elevation²	0.13	0.14	−0.15, 0.40

Table S7.B. Predicted elevational optima and range shift between historical (1976–2000) and present (2025) surveys for the model without relocation-error-informed priors. Estimates represent posterior medians, with posterior standard deviations (SDs) and 95% credible intervals (CrIs) reported for each optimum and the estimated range shift.

Measure	**Posterior median (m)**	**Posterior SD (m)**	**95% CrI (m)**
Historical optimum	1304	58	1196–1426
Present optimum	1360	69	1239–1516
Range shift	+56	31	+9 to +128

Table S7.C. MCMC convergence and sampling diagnostics for the full Bayesian model without informative priors. All parameters had Rhat values of 1.00, indicating convergence among the four MCMC chains. Effective sample sizes were high across the fixed effects, with all Bulk ESS values and Tail ESS values >1000. No divergent transitions were detected during NUTS sampling.

Diagnostic	**Result**
MCMC chains	4
Iterations per chain	4,000
Warmup iterations per chain	1,000
Post-warmup draws	12,000
Rhat	1.00 for all parameters
Bulk ESS, fixed effects	>1000 for all parameters
Tail ESS, fixed effects	>1000 for all parameters
Divergent transitions	0

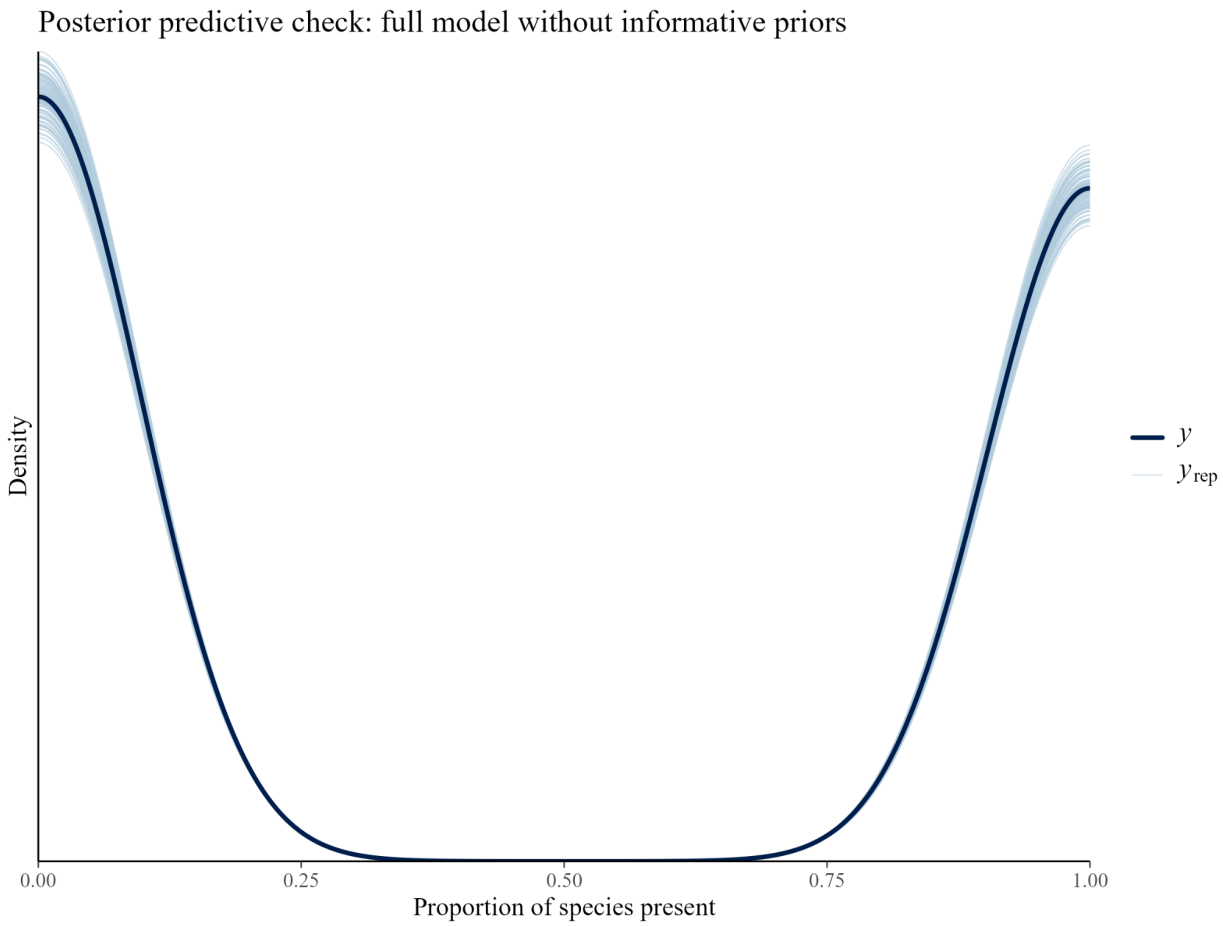


Figure S7.B. Posterior predictive check of the full model without informative priors. The consistency of the real data and simulated data demonstrate that the model accurately characterizes the data in this density graph. The simulated data is shown by the dark blue curve (y) and observed data is shown by the light blue curves (y_{rep}, n = 100).

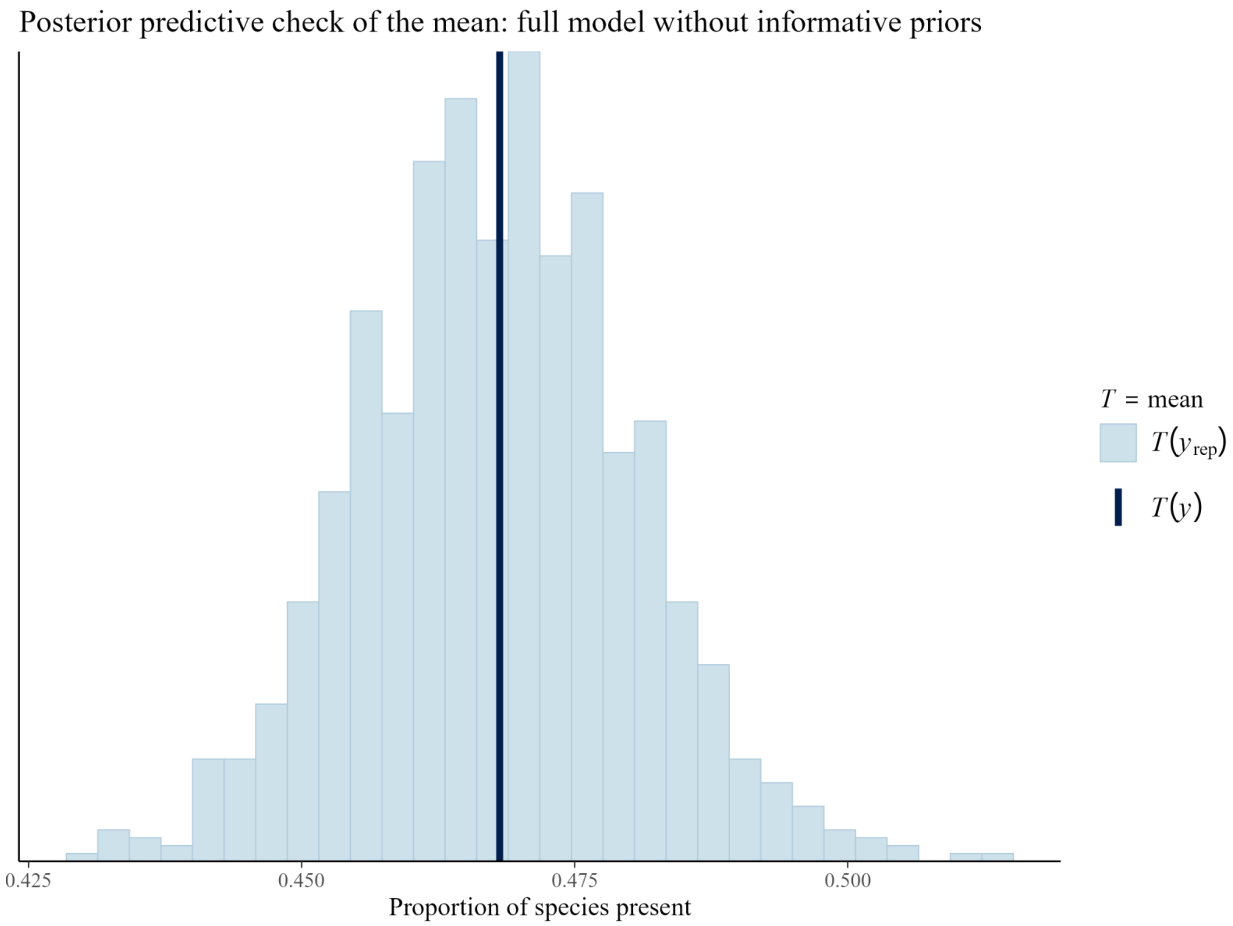


Figure S7.C. Posterior predictive check of the full model with no informative priors. The consistency of the real data and simulated data demonstrate that the model accurately characterizes the mean proportion of species present in this histogram. The simulated data is shown by the dark blue line (y) and observed data is shown by the light blue (y_{rep}, n = 1000).

S8. Model diagnostics of the relocation error model for extracting informative priors

Table S8.A. Posterior estimates for the relocation error Bayesian model. Posterior means, standard errors, and 95% credible intervals are shown for each regression coefficient. Bolded credible intervals exclude 0.

Parameter	Estimate	Est. Error	95% CrI
Intercept	−1.89	1.74	−5.32, 1.64
Treatment: Error	0.72	2.24	−3.72, 5.26
Elevation (linear)	−4.90	2.20	**−9.51, −0.83**
Elevation²	−1.68	1.61	−5.01, 1.35
Latitude	0.96	1.64	−2.21, 4.33
Treatment × Elevation	0.45	1.41	−2.35, 3.27
Treatment × Elevation²	−0.18	1.91	−4.08, 3.61

Table S8.B. MCMC convergence and sampling diagnostics for the relocation error Bayesian model. All parameters had Rhat values of 1.00, indicating convergence among the four MCMC chains. Effective sample sizes were high across the fixed effects, with Bulk ESS values ranging from 5,477 to 6,638 and Tail ESS values ranging from 6,296 to 7,143. No divergent transitions were detected during NUTS sampling.

Diagnostic	Result
MCMC chains	4
Iterations per chain	4,000
Warmup iterations per chain	1,000
Post-warmup draws	12,000
Rhat	1.00 for all parameters
Bulk ESS, fixed effects	>1000 for all parameters
Tail ESS, fixed effects	>1000 for all parameters
Divergent transitions	0

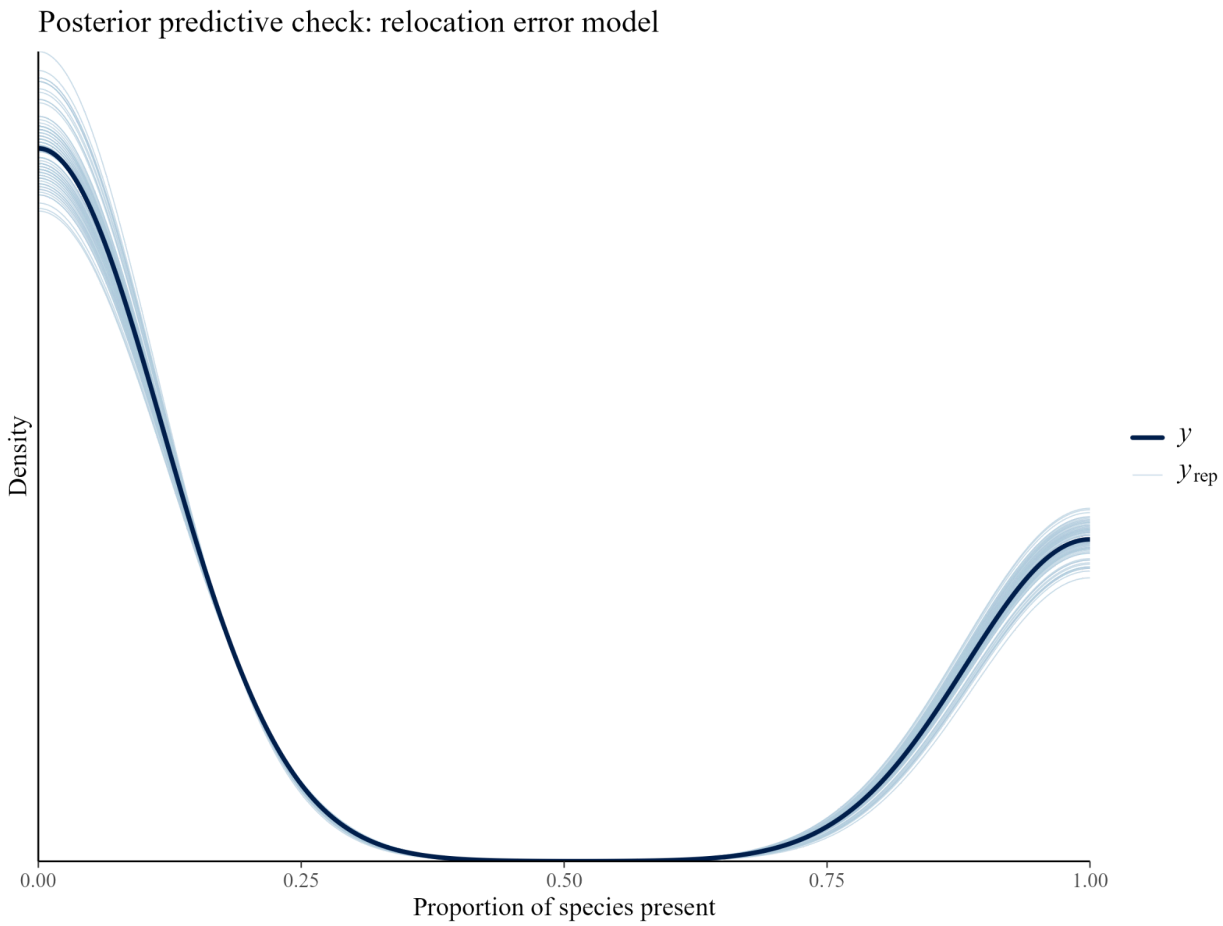


Figure S8.A. Posterior predictive check of the relocation error model. The consistency of the real data and simulated data demonstrate that the model accurately characterizes the data in this density graph. The simulated data is shown by the dark blue curve (y) and observed data is shown by the light blue curves (y_{rep}, n = 100).

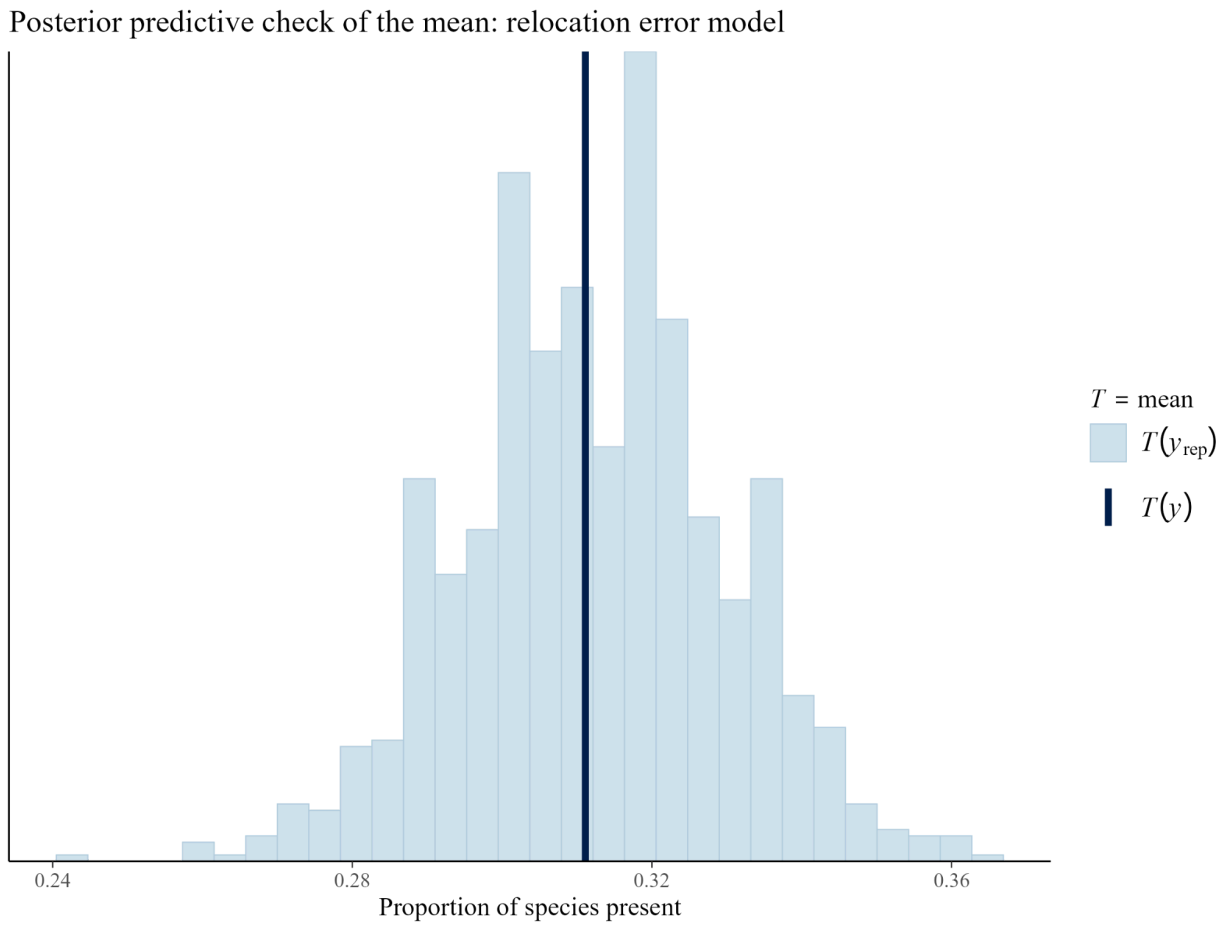


Figure S8.B. Posterior predictive check of the relocation error model. The consistency of the real data and simulated data demonstrate that the model accurately characterizes the mean proportion of species present in this histogram. The simulated data is shown by the dark blue line (y) and observed data is shown by the light blue (y_{rep}, n = 1000).

S9. Informative priors used in the relocation-error-informed Bayesian model with the functional group interaction

Table S9. Informative priors used in the Bayesian model with the functional group interaction.

Parameter	**Prior**	**μ**	**SD**
Time: Present	Normal	0.50	2.73
Time × Elevation	Normal	0.76	1.76
Time × Elevation²	Normal	0.09	2.35
Time × Shrub	Normal	1.07	1.74
Time × Tree	Normal	−0.23	2.02
Time × Elevation × Shrub	Normal	−1.62	1.66
Time × Elevation² × Shrub	Normal	−1.53	1.91
Time × Elevation × Tree	Normal	0.28	1.44
Time × Elevation² × Tree	Normal	0.25	1.90

S10. Model diagnostics of the full Bayesian model with relocation-error-informed priors and the functional group interaction

Table S10.A. Posterior estimates for the Bayesian model including functional group interactions. Posterior means, standard errors, and 95% credible intervals are shown for each regression coefficient. Herb species were the reference functional group; therefore, shrub and tree coefficients represent differences relative to herbs. Bolded credible intervals exclude 0.

Parameter	Estimate	Est. Error	95% CrI
Intercept	0.46	0.43	−0.38, 1.31
Time: Present	0.46	0.25	−0.03, 0.94
Elevation (linear)	−0.29	0.61	−1.51, 0.92
Elevation²	−1.59	0.34	**−2.27, −0.93**
Functional Group: Shrub	−0.03	0.54	−1.09, 1.06
Functional Group: Tree	0.51	0.68	−0.81, 1.88
Latitude	0.82	0.22	**0.41, 1.26**
Time × Elevation	0.81	0.26	**0.31, 1.32**
Time × Elevation²	0.64	0.24	**0.17, 1.11**
Time × Shrub	0.25	0.34	−0.42, 0.93
Time × Tree	−0.23	0.44	−1.10, 0.62
Elevation × Shrub	0.36	0.79	−1.20, 1.91
Elevation² × Shrub	0.29	0.40	−0.49, 1.09
Elevation × Tree	−0.42	0.98	−2.41, 1.50
Elevation² × Tree	0.69	0.49	−0.28, 1.64
Time × Elevation × Shrub	−0.75	0.33	**−1.40, −0.11**
Time × Elevation² × Shrub	−0.63	0.32	**−1.24, −0.01**
Time × Elevation × Tree	−0.69	0.40	−1.49, 0.10
Time × Elevation² × Tree	−0.99	0.39	**−1.74, −0.24**

Table S10.B. MCMC convergence and sampling diagnostics for the full model with the functional group interaction. All parameters had Rhat values of 1.00, indicating convergence among the four MCMC chains. Effective sample sizes were high across the fixed effects, with Bulk ESS values ranging from 3,356 to 10,122 and Tail ESS values ranging from 5,163 to 9,638. No divergent transitions were detected during NUTS sampling.

Diagnostic	**Result**
MCMC chains	4
Iterations per chain	4,000
Warmup iterations per chain	1,000
Post-warmup draws	12,000
Rhat	1.00 for all parameters
Bulk ESS range, fixed effects	>1000 for all parameters
Tail ESS range, fixed effects	>1000 for all parameters
Divergent transitions	0

Table S.10.C. Bayesian R^2 for Bayesian models with functional groups.

Model	**Bayesian R^2**	**95% CrI**
Full model	0.398	0.375–0.417
Relocation-error model	0.628	0.581–0.668
Full model without informative priors	0.398	0.375–0.418

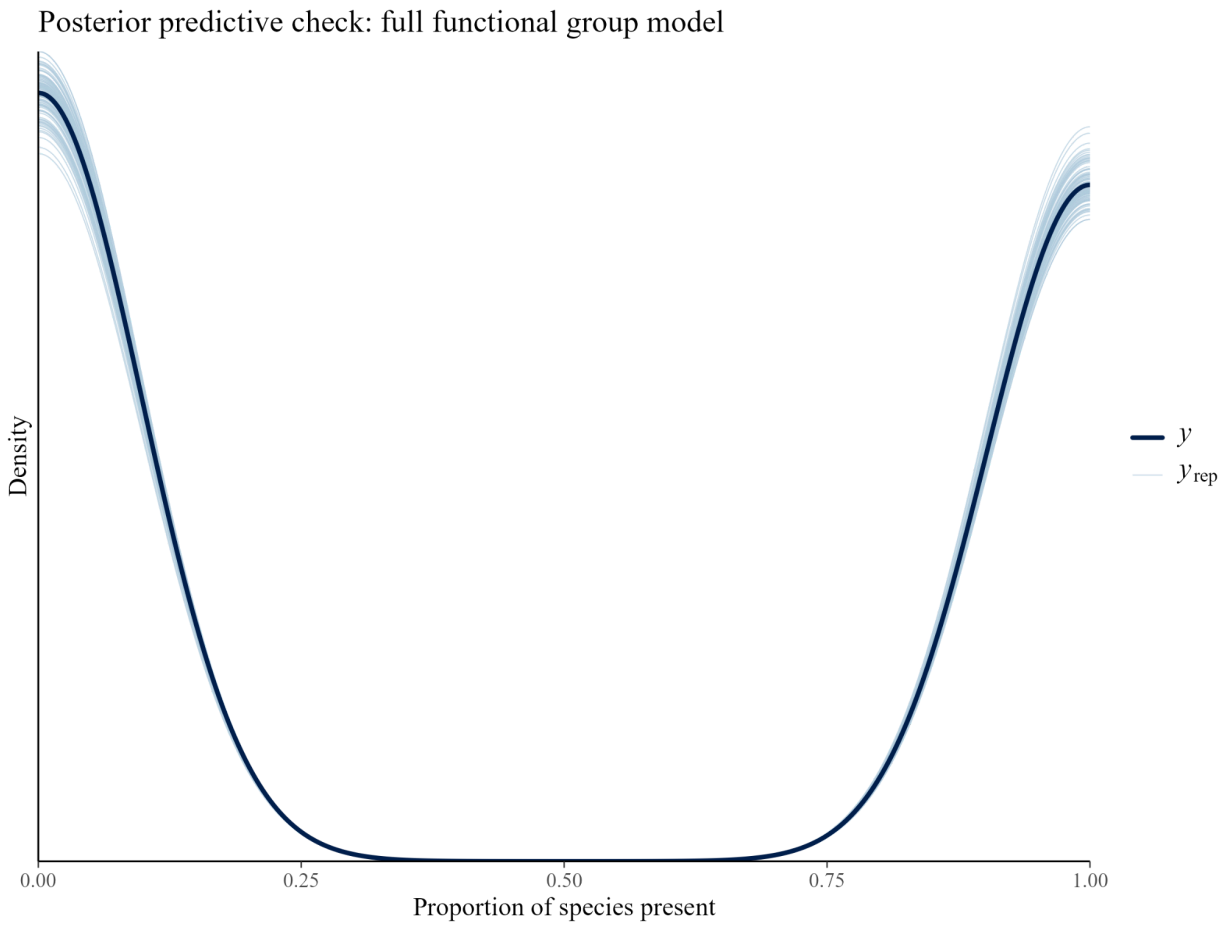


Figure S10.A. Posterior predictive check of the full model with the functional group interaction and informative priors. The consistency of the real data and simulated data demonstrate that the model accurately characterizes the data in this density graph. The simulated data is shown by the dark blue curve (y) and observed data is shown by the light blue curves (y_{rep}, n = 100).

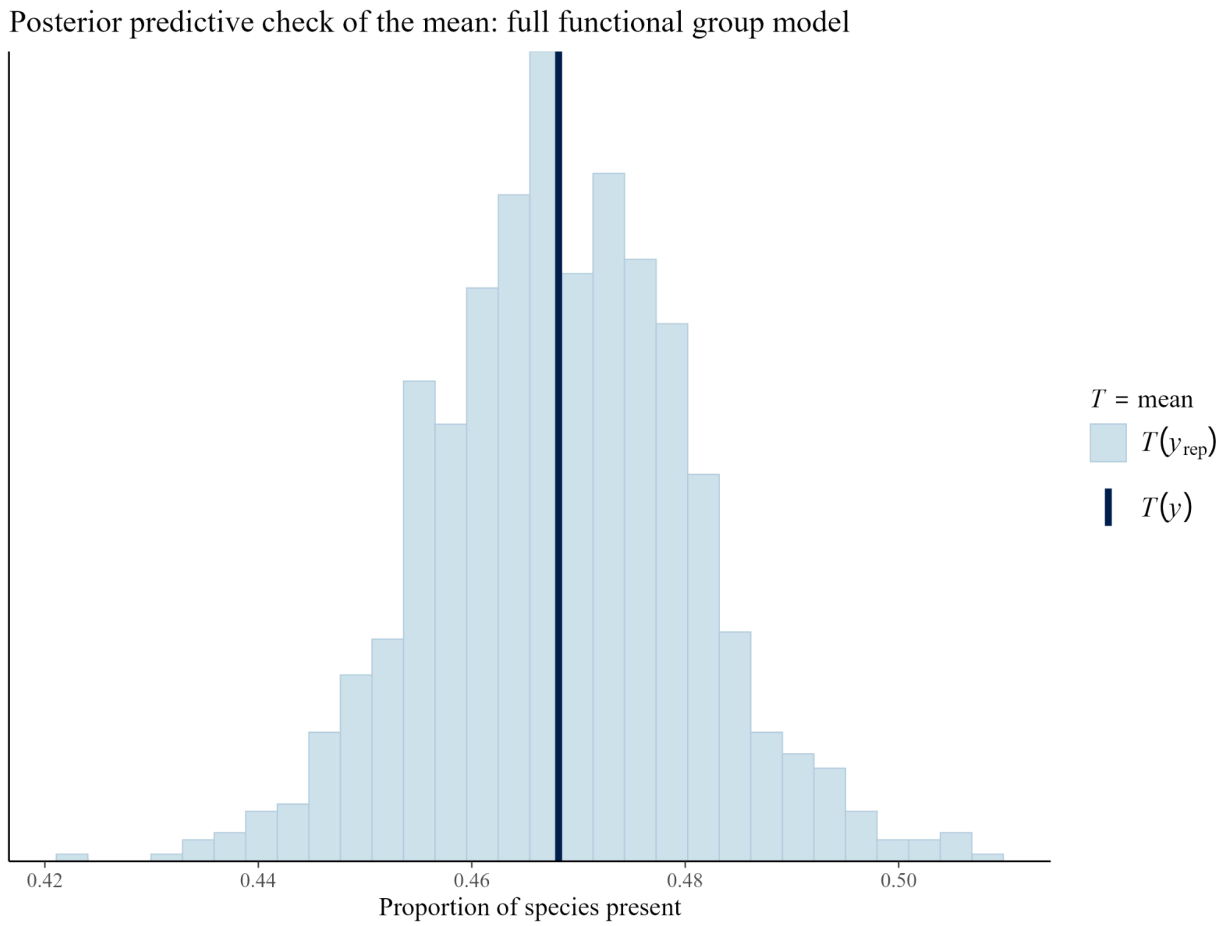


Figure S10.B. Posterior predictive check of the full model with the functional group interaction and informative priors. The consistency of the real data and simulated data demonstrate that the model accurately characterizes the mean proportion of species present in this histogram. The simulated data is shown by the dark blue line (y) and observed data is shown by the light blue (y_{rep}, n = 1000).

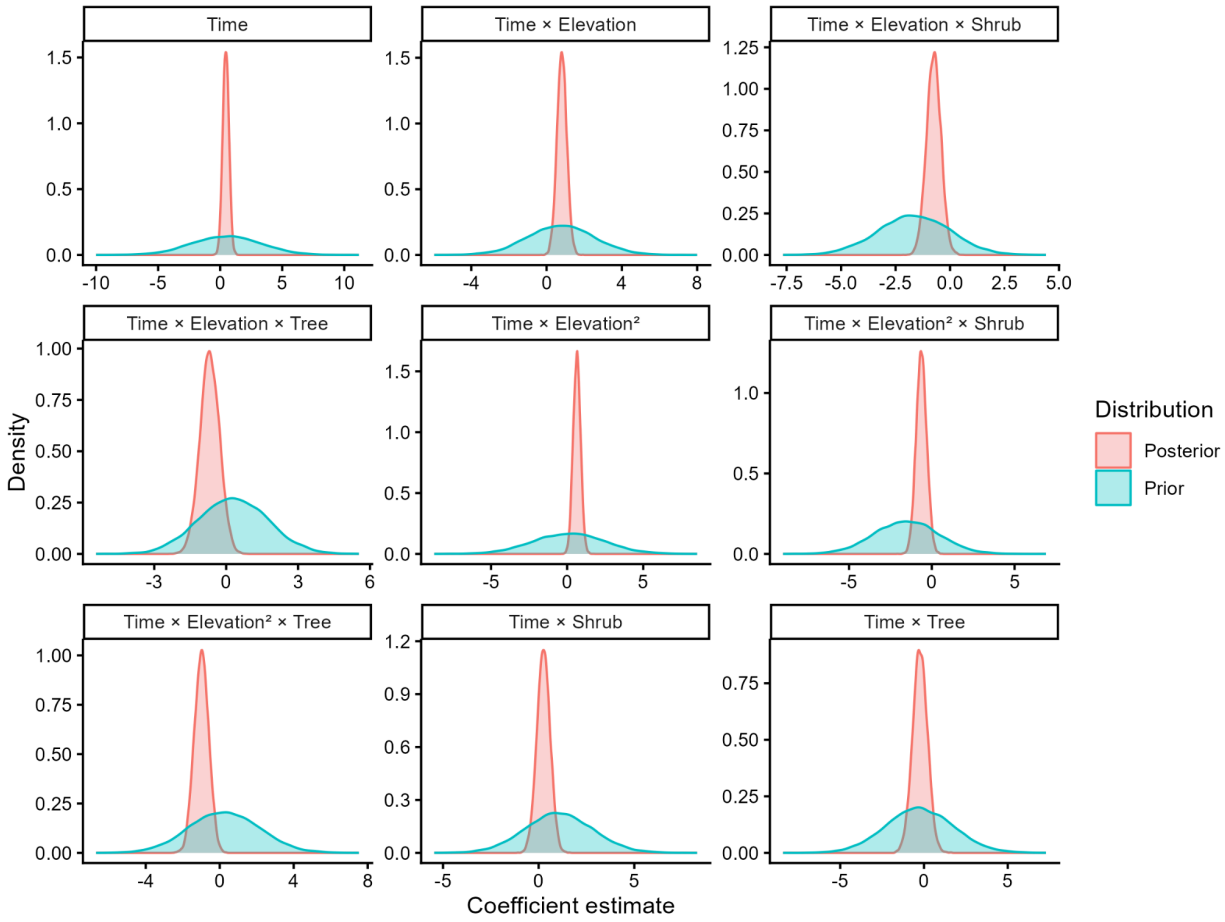


Figure S10.C. Prior (blue) and posterior (red) distributions for fixed-effect coefficients in the Bayesian model. Coefficients include time, time × elevation, time × elevation², and interactions between time and functional group (shrub and tree), including their interactions with elevation and elevation².